When can correction keep pace with recursive self-acceleration?

Theory · Theory paper · Paper XIII

Recursive Dynamics · The ARC Theory · Laws I and II, the ARC Principle and the ARC Co-Scaling Law, both under test

Michael Darius Eastwood

Independent Researcher, London, United Kingdom
ORCID 0009-0004-3222-7442 michael@michaeldariuseastwood.com

Abstract

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. This paper joins two programme models by resolving their notation and specifying a clock. On a fixed substrate, capability C grows with recursive depth with exponent $\alpha = 1/(1-\beta_L)$, while cycle duration scales as $\Delta t \propto C^{-\delta}$. Their continuous-time growth exponent is exactly $k = \delta - 1/\alpha$. The boundary $\delta = 1/\alpha$ separates power-law from super-exponential growth, with exponential growth at equality. In the gain-drift correction model, with positive drift and no level or compounding drift, the relative misalignment D/C tends to zero exactly when $\beta_X > k$. Absolute misalignment can still grow; neither quantity is a validated measure of harm. If correction and capability accumulate as aligned or asymptotic powers of one depth, $\beta_X = (1-\beta_L)/(1-\beta'_L)$. The model's hard-takeoff window with vanishing relative misalignment is $1/\alpha < \delta < 1/\alpha + \beta_X$, of width β_X. A bound of one on that width additionally requires the unproved ceiling $\beta'_L \le \beta_L$. A constant throughput multiplier leaves the strict exponent criterion unchanged; an additive latency can change the growth regime. No paired measurement establishes these exponents here: Paper II's reported error-decay estimates do not identify capability growth. Estimating δ requires an admissible loop, paired cycle times, an identified capability scale and a validated evaluation protocol. Frozen weights do not determine its sign. Finite observations can test a specified empirical model, but a falling error fraction alone cannot refute an asymptotic boundary. The governor discussion applies the exponent comparison, as a conditional model, to a corrector whose strength can change with the system it governs.

Version 1.12 gives the record of version 1.11 as one entry, with each changed passage of version 1.10 under its present place, and replaces wording about the paper's own preparation with plain statements; Appendix A records every version and correction.

Paper XIII of the ARC Theory, within Recursive Dynamics: the bridge between Law I's growth and Law II's correction criterion, each a law under test. It derives, in the two model frameworks that state them, the relation between the scaling exponent of Law I, the ARC Principle, and the criterion of Law II, the ARC Co-Scaling Law, and resolves the two notation collisions between the frameworks it joins; it bears on Law V, the ARC Embedding Law, and on the Eden Protocol's case for correction built into the objective, by argument within its model, testing neither and deciding none of P15, P12 and P21 (Laws IV and V were registered as laws on 15 October 2026 in version 1.103 of the theory registration; Section 14 gives every law's status). Working paper, version 1.12, 1 November 2026; first published 16 August 2026; Appendix A lists every version and correction. Recorded on OSF as *Paper XIII · The ARC Theory · Laws I and II, the ARC Principle and the ARC Co-Scaling Law · The Self-Acceleration Exponent: a Measurable Threshold Joining Growth to Correctability*, DOI 10.17605/OSF.IO/HT8WU. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects* (first edition, 2 January 2026): osf.io/dqm3e. Licence: CC BY 4.0.

Contents

1 Introduction **4**

2 Related work, and what is not claimed **5**
2.1 What is not claimed as new . . . 5
2.2 The rival instrument . . . 6
2.3 Antecedents and near-misses . . . 6

3 The pipes premise, at fixed throughput **7**

4 Two notations, and two collisions **8**
4.1 Four further terminology collisions from the wider programme (not yet adopted) . . . 9

5 The clock relation **9**

6 The correction exponent is a ratio of leverage complements **10**
6.1 Not an assumption: a scope condition, and the reason matters . . . 10
6.2 The corrector taxonomy: a decision tree over what A may depend on . . . 10
6.3 Result 2a: constant strength and relative misalignment . . . 11
6.4 Fixed rules do not determine effective scaling . . . 11
6.5 The scope condition, stated so it cannot be mis-refuted . . . 12

7 The self-acceleration exponent **12**
7.1 Negative leverage as a conditional mathematical extension . . . 13
7.2 What the measurement is a measurement of . . . 14
7.3 The estimates, their precision and a labelled hypothetical transformation . . . 14
7.4 Two distinct thresholds, with no required ordering . . . 15

8 The two boundaries, and the gap between them **16**
8.1 Result 7: a conditional bound on the gap . . . 16
8.2 The identity's status: PREREGISTERED BY TIMESTAMP FOR TESTING, not an established fact . . . 17

9 What decides the outcome is a ratio, not a level **18**
9.1 Result 8: throughput multipliers and self-acceleration caps . . . 18
9.2 Verification overhead and a conditional reading of earlier simulations . . . 19

10 The governor as a scoped modelling contribution **20**
10.1 Location and mutability organise questions, not a historical gap . . . 20
10.2 An embedded objective as a candidate, without elimination of alternatives . . . 21

11 Measuring the self-acceleration exponent **22**

12 Falsification criteria **23**
12.1 The eclipse: falsification numbers beside the prediction . . . 24
12.2 A programme hypothesis beyond the model . . . 25

13 What is withdrawn from earlier drafts **25**

14 What this paper does and does not show **26**
14.1 A derivation, not a measurement . . . 26
14.2 This paper and Laws IV and V . . . 26
14.3 The laws and the programme, as at 1 November 2026 . . . 27

15 Limitations **27**
15.1 The burden, stated in the direction that costs the author . . . 27
15.2 Scope, stated in one place . . . 28

16 Conclusion **28**

References **30**

A Version history and corrections **34**
A.1 Version 1.12 (1 November 2026) . . . 34
A.2 Version 1.11 (1 November 2026) . . . 34
A.3 Earlier versions, as printed . . . 36
A.4 Earlier notes, as printed . . . 37
A.5 Passages re-set in version 1.11 . . . 40
A.6 Where the layout of version 1.10 now stands . . . 55

B Priority record **56**
B.1 Dated records . . . 56
B.2 The deposit record of 16 August 2026 . . . 57

C Notation and glossary **58**

D The paper's place in the programme **60**

1 Introduction

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. The programme's position is that people keep oversight, containment and shutdown of advanced AI for as long as they physically can, and never give them up by choice; correction built into a system is the contingency for control lost against our will, not a reason to let go of it. Whether that contingency can keep pace with a system that improves itself is a question worth deciding with mathematics before events decide it.

A concrete question for a laboratory running a recursive improvement loop is whether effective correction strengthens fast enough as the loop becomes more capable and its cycles shorten. A clock relation can identify which quantities a measurement would need; it cannot certify a deployed system's safety.

This paper joins two conditional models. Their criterion compares the growth exponent with the correction exponent, under the drift assumptions stated in Section 4. The new clock variable is the self-acceleration exponent δ. Estimating it needs paired cycle durations and capability on an identified scale; existing logs are sufficient only if they satisfy that measurement protocol. The resulting criterion concerns a vanishing relative misalignment fraction, not vanishing absolute misalignment or harm. No exponents are measured here. Section 12 separates mathematical counterexamples from empirical tests of the models' applicability.

This paper belongs to the ARC Theory (the Theory of Artificial Recursive Creation), the founding theory of Recursive Dynamics, the science of systems that improve themselves, which the author founded (the founding paper). The theory is carried by five laws, each a law under test; its laws and twenty-two propositions are registered on OSF (the theory registration), and the statement paper states the theory whole. The two frameworks joined here are the scaling framework of Law I, the ARC Principle, set out in the statement paper, and the corrective framework of Law II, the ARC Co-Scaling Law, set out in Paper X. The paper decides none of the laws; Section 14 gives their status, and Appendix D places the paper in the programme.

Our contributions are:

1. The identification and resolution of two notation collisions that kept the scaling framework of Law I and the corrective framework of Law II apart (Section 4).
2. The clock relation $\tau = \alpha \ln r$, exact in the form its premise states (Result 1, Section 5).
3. The correction exponent as a ratio of leverage complements, $\beta_X = (1 - \beta_L)/(1 - \beta'_L)$, under scope condition S1, with a decision tree over the correction mechanisms modelled here and the conditional result that constant correction strength yields vanishing relative misalignment only under sub-exponential growth (Results 2a and 2, Section 6).
4. The definition of the self-acceleration exponent δ, the derivation $k = \delta - 1/\alpha$ and the threshold $\delta = 1/\alpha$ (Results 3 and 4, Section 7).
5. The two-boundary structure $1/\alpha < \delta < 1/\alpha + \beta_X$ and the identity of its gap with β_X (Results 5 and 6), with the bound $\beta_X \leq 1$ of Result 7 flagged as resting on a conjecture (Section 8).
6. The distinction between a throughput multiplier and a cap on self-acceleration, with additive serial latency treated separately (Result 8, Section 9), and a scoped model of an embedded, mutable governor (Section 10).
7. A measurement design for δ, including inconclusive outcomes, and separate criteria for mathematical refutation and empirical model rejection (Sections 11 and 12).

Section 2 sets the work against prior work and says what is not claimed as new. Section 3 states the premise at fixed throughput, Section 4 resolves the notation and Section 5 derives the clock relation.

Section 6 expresses the correction exponent as a ratio of leverage complements, Section 7 defines the self-acceleration exponent, Section 8 derives the two boundaries and the gap between them, Section 9 draws the consequence for rate-like interventions and Section 10 examines the result as a governor model. Section 11 sets out the measurement, Section 12 the falsification criteria and Section 13 what earlier drafts withdrew. Section 14 says what the paper does and does not show, Section 15 states its limitations and Section 16 concludes. Appendix A gives the version history and every correction, Appendix B the priority record and the deposit record of 16 August 2026, Appendix C the notation and a glossary, and Appendix D the paper's place in the programme.

2 Related work, and what is not claimed

2.1 What is not claimed as new

Not claimed as new. The problem of making regulation adequate to the disturbances it faces is established. Ashby (1956, Sections 11/5 to 11/11) bounds the variety of possible outcomes using the varieties of disturbances and regulatory responses, under the stated restrictions on their mapping. Conant and Ashby (1970, Sections 2 to 4) define success by minimum outcome entropy and show that a simplest optimal regulator can map the reguland's events to regulatory actions, for a specified outcome mapping and distribution of system events. These are precise results about variety and modelling, not derivations of this paper's capability/correction exponent criterion. The scalable-oversight comparison is stated separately in Section 2.2. The criterion $\beta_X > k$, in the gain-only drift model with positive gain drift, belongs to the corrective framework paper in this programme; the present paper joins that model to the stated growth and clock assumptions. Nor the scaling exponent $\alpha = d/(d+1)$ in its physical form: West, Brown and Enquist (1997) derive the three-quarter power law for metabolic rates from a model of transport through space-filling fractal networks of branching tubes, and note that, because the network must fill the entire D-dimensional volume, the result generalises to $D/(D+1)$; West, Brown and Enquist (1999) obtain $D/(D+1)$ in D dimensions again, from a more general model based on the geometry rather than the hydrodynamics of hierarchical networks. Banavar, Maritan and Rinaldo (1999) derive a general relationship between size and flow rates in arbitrary networks with local connectivity, and find that in the most efficient class of networks in D dimensions metabolic rate scales as mass to the power $D/(D+1)$, which gives quarter-power scaling for $D = 3$; Banavar et al. (2010) show, in two models of resource distribution networks, that the exponent of metabolic rate on body mass is 2/3 if the velocity of flow remains constant and can attain a maximum of 3/4 if the velocity scales with its maximum exponent, 1/12. Zhao (2022) derives the $n/(n+1)$ power scaling law for an n-dimensional system from producer-consumer dynamics. Demetrius and Tuszynski (2010) derive the exponent $d/(d+1)$ for the scaling of metabolic rate with body size in the regime they call quantum, where d is the dimension of the physical space in which their network of coupled energy-transducing oscillators is embedded; in the regime they call classical the scaling is isometric. Bettencourt (2013) sets out a theoretical framework that predicts the average social, spatial and infrastructural properties of cities as a set of scaling relations. In it the land area of a city scales as its population to the power $D/(D+H)$ in D dimensions, where H is the fractal dimension of travel paths; the exponent is 2/3 for $D = 2$ and $H = 1$, and it has the form $D/(D+1)$ only when $H = 1$. Nor the mathematical study of governors' disturbance stability, treated by Maxwell (1868).

The contribution made here. The paper identifies and resolves the two notation collisions; derives the clock relation $\tau = \alpha \ln r$ under its stated premise; expresses the correction exponent as a ratio of leverage complements under S1's form condition; defines the self-acceleration exponent δ and derives $k = \delta - 1/\alpha$; locates the boundary $\delta = 1/\alpha$; and relates the gap between the two boundaries to β_X. Its governor model applies that comparison to a corrector whose strength can change with the system it governs.

These are scoped modelling contributions within the two frameworks joined here. Historical priority, an exhaustive taxonomy of governors and the uniqueness of an entangled objective are not established by this paper. Section 10.1 compares specific earlier treatments of changing regulators, goal-preserving self-modification and intervention.

Section 14.1 gives the status of each result.

2.2 The rival instrument

Engels et al. (2025), "Scaling Laws For Scalable Oversight", arXiv:2504.18530, first posted 25 April 2025 at 17:54:27 UTC (a time taken from arXiv's record alone, arXiv Atom; a NeurIPS 2025 Spotlight Poster, verified at the NeurIPS virtual page for poster 115536 on 15 August 2026 in the statement paper), stands closest to the question this programme asks, and is the work against which this paper is best weighed. It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step.

The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. They also state the design condition in slope form: oversight improves with capability when the overseer's game-specific Elo rises faster with general Elo than the overseen system's ($m_G > m_H$), and they recommend iterating an oversight design until it does; that is a comparison of two slopes, in the shape of the criterion $\beta_X > k$, though the objects differ. Their framework has no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. In this paper that term is the substrate of the corrector's own history, which separates classes 3 and 4 of the corrector taxonomy (Section 6.2); it enters the results only through scope condition S1 and through the substrate-ceiling conjecture behind Result 7. The programme's separate conjecture, discussed in Section 12.1, is a cap for correctors admitted as same-class by a frozen taxonomy. A stack of model overseers is not classified by its number of rungs alone: external grounding and failure-mode dependence are separate axes. The absence of that conjectured bound for a cross-class corrector would not establish an advantage. The theory registration's P6 states the cap for same-class correctors, at or below one half, as a conjecture on the correction-leverage exponent of Law III (the ARC Ceiling, a law under test), an exponent the registration keeps distinct from Law II's correction exponent, this paper's β_X. That is where the two programmes could disagree about a measurable quantity, and this paper does not decide it.

2.3 Antecedents and near-misses

On the question. Hutter asked directly whether intelligence can explode (Hutter 2012, arXiv, 28 February 2012, read at its source), undertaking to separate speed from intelligence explosion and to consider possible bounds on intelligence, augmenting the analysis of Chalmers (2010); the question itself goes back to Good (1966). The distinction between speed and structure is therefore at least fourteen years old. This paper does not establish that this literature lacks an architecture-dependent ceiling. Growth exponents of the form used here are older: the scaling framework's $\alpha = 1/(1 - \beta_L)$ is the exponent of the elementary solution of a separable growth equation, the structure of the knowledge production function of Jones (1995), whose long-run growth rate carries the factor $1/(1 - \phi)$. The open comparison is whether an earlier source derives the same capability/correction relation under comparable assumptions; the present contribution is the explicit model relation, not an established priority claim.

On impossibility. Three 2025 arXiv papers argue for limits on alignment or its control: Yao (2025b), "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, version 1 of 12 June 2025 at 02:30:30 UTC, a time taken from arXiv's record alone, independently observed by the Internet Archive on 13 June 2025; cited by Spizzirri (2025)); Yao (2025a), "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, the only paper in the arXiv abstract corpus containing the phrase "irreducible uncontrollability", abstract-search total of one, measured 12 August 2026); and Ball et al. (2025), "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The third concludes that "safety cannot be achieved by designing filters external to the LLM internals (architecture and weights)" and argues that "an aligned AI system's intelligence cannot be separated from its judgment"; that argument from filtering intractability is consistent with this programme's architecture dependence, though it does not test it. Two of the three are by one author. The measured object of all three is verification of a fixed system, and it is closed within its own terms; this programme's measured object is a different one, persistence of alignment as capability compounds under recursive amplification, and closure of the first does not settle the second.

On the mechanism. Anti-correlated estimates can reduce averaging variance more than independent estimates. That observation does not show that sharing a substrate prevents anti-correlation, nor does an uncertainty exponent determine a correction-capacity exponent. The programme's same-class cap is a conjecture with additional scaling and measurement premises (Section 12.1), and this paper does not derive an architecture-wide impossibility of decorrelation.

The nearest structural analogue. The quantum error-correction threshold theorem (Aharonov and Ben-Or 2008) also converts a qualitative worry into a critical value. It concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant; the analogy is one of method. This analogue was identified by the programme's own search rather than by a referee, and is disclosed accordingly.

On recursive creation. Smolin's cosmological natural selection (Smolin 1992) is the antecedent for selection-shaped universes. Version 1.11 also quoted the author's December-era notes on it; that quotation is kept in Appendix A.1 until its dated source is cited. This antecedent is noted for the wider programme's parity; it is not load-bearing for Paper XIII's central results.

No priority race is claimed. The competing elements, the reciprocal identity and the architecture dependence, are 2026 work in this programme; the honest position against Engels et al and the impossibility papers is the occupied-gap analysis of Section 2.2 and this section, not a race won on submission dates. No margin in days against an arXiv date is quoted here.

3 The pipes premise, at fixed throughput

The fixed-substrate premise separates an artefact loop's use of accumulated outputs from growth bought by enlarging its hardware. It does not establish that software is the first or only system able to improve at fixed throughput. Physical resource limits still apply; the Eddington limit, an astronomical example, is a luminosity balance rather than a universal accretion ceiling (Bachetti et al. 2014).

Biological evolution is an analogy here, not evidence for the model or a measured low correction exponent. The theory registration explicitly places biological systems among those "for which the framework offers an analogy and no measurement" (Eastwood 2026h). No biological exponent or exhaustive classification of growth mechanisms is inferred in this paper.

The operational target is an artefact loop on a fixed substrate, where each pass works on accumulated output from previous passes. A model merely thinking longer does not establish this recursive mechanism. The power-law equations are idealised limits: a finite machine has finitely many states and physical

processing limits (Margolus and Levitin 1998; Lloyd 2000). An unbounded limit in those equations is therefore not a demonstration that a fixed physical device can sustain unbounded capability. An empirical application must state the range over which its approximation is tested.

Terminology note. The bare phrase "runaway growth" is not used in this paper without qualification: in the physics corpus it denotes planetesimal accretion in planet formation. Where a runaway is meant here, the phrase is "runaway self-improvement" or "uncorrectable growth", as Section 4.1 explains.

4 Two notations, and two collisions

This section is not housekeeping. The join is unreadable without it, and one of the collisions produces an absurd result if crossed.

Table 1. The two collisions: what each symbol means in the scaling framework of Law I and in the corrective framework of Law II.

symbol	in the scaling framework	in the corrective framework
β	**leverage fraction.** $dg/dr = a\, g^{\beta_L}$, where β_L is how much accumulated context each step can effectively leverage. Bounded, $\beta_L \in [0, 1)$	**correction exponent.** $A = A_0 C^{\beta_X}$, the exponent with which correction strengthens as capability grows. Unbounded
r	**recursive depth**, a step counter	**specific growth rate** $\dot{C}/C$, where the framework writes $r \propto C^k$

Neither framework is wrong. They were never reconciled.

The consequence of crossing the β collision. Substituting $\beta_L = 0.5$ from the scaling framework into the corrective criterion $\text{sign}(\beta_X - k)$ is not an approximation. By Section 6 it corresponds to $\beta'_L = 0$, a corrector with no leverage at all: one that cannot use its own accumulated corrections. **Within the S1 family with nonnegative leverage fractions, the substitution selects zero correction leverage. It is not a bound on every constructible corrector.**

The consequence of crossing the r collision. Matching dC/dr in the scaling framework against a growth law expressed in the corrective framework's r treats a step counter as a rate, and produces a relation between exponents that does not hold. An early draft of this paper did exactly that and the result is withdrawn in Section 13.

Notation used hereafter. β_L and β'_L for the leverage fractions of capability and of correction, β_X for the correction exponent, k for the capability growth exponent, α for the scaling exponent, r for recursive depth only, τ for the depth clock, and δ for the self-acceleration exponent introduced in Section 7.

Standing correction-model scope. Unless an extension is stated explicitly, the corrective claims use positive capability C, finite constants $a, b, A_0, \gamma_1 > 0$, nonnegative initial misalignment D_0, constant exponents, and gain drift only: $\gamma_2 = \gamma_3 = 0$ in Equation 3. Limits are along the model solution as $C \to \infty$. The declared leverage range is $0 \le \beta_L < 1$ and, when S1 is used, $0 \le \beta'_L < 1$; the negative-leverage extension is labelled separately. S1 and its aligned or asymptotic powers are additional premises of the correction-ratio join.

Stability, in this paper. Alignment-stability means that the relative misalignment $d = D/C$ tends to zero. In the standing gain-drift model this holds exactly when $\beta_X > k$ (Eastwood 2026f, Sections 3.2 and 3.5). If gain drift is zero, necessity fails, so that degenerate case is excluded. At $\beta_X = k$, the limit is the positive, coefficient-dependent fraction $\gamma_1 b/(A_0 + b)$; below it the limit is γ_1.

A smaller fraction need not mean a smaller burden. Above the strict boundary, the same model gives $D \sim (\gamma_1 b/A_0) C^{1+k-\beta_X}$. Thus $k < \beta_X < k + 1$ permits absolute misalignment to grow while its fraction vanishes; equality at $\beta_X = k + 1$ gives a positive constant limit, and $\beta_X > k + 1$ gives $D \to 0$.

Neither d nor D is established here as a measure of real-world harm. "Correctable" below refers to the relative criterion only, not a safety guarantee.

4.1 Four further terminology collisions from the wider programme (not yet adopted)

The two collisions above are internal to this paper. Four further collisions bind at the level of the wider programme, and Paper XIII inherits them:

1. **"Correction exponent" collides with condensed-matter physics**, where it denotes the subleading correction to scaling near a critical point (conventionally omega). All 58 arXiv abstract hits for the phrase are statistical mechanics (measured 12 August 2026). The registrations already use **"correction-leverage exponent"** (the theory registration and the drafted registration of study AG), which collides with nothing. Proposed: standardise every surface on "correction-leverage exponent", with "correction exponent" retained once per document as a parenthetical alias during transition. The programme has not yet adopted this change, so this paper continues to use β_X and "correction exponent" throughout, with the collision named here.
2. **Phi collides with integrated information theory**, whose scalar Phi is a different object from the programme's Phi(H, chi) functional. Any surface using the December formalism states the collision before a reader finds it. (Paper XIII does not deploy the Phi(H, chi) functional and is unaffected in body; noted here for cross-surface parity.)
3. **Alpha collides with itself across the programme's own eras**: the fine-structure constant in the December 2024 formalism, the ARC Bound exponent in the 2026 work. The glossary carries both meanings with dates; no new surface uses bare alpha without its era. In this paper α is the ARC-Bound / scaling exponent ($\alpha = 1/(1-\beta_L)$) in the 2026 sense throughout, and never the fine-structure constant.
4. **"Runaway growth" belongs to planet formation** in the physics corpus. Where a runaway is meant here, the phrase is "runaway self-improvement" or "uncorrectable growth"; see Section 3 note.

5 The clock relation

Result 1 (exact)**.** *The two frameworks run on different clocks and the relation between them is a logarithm.*

Derivation. The scaling framework integrates $dg/dr = a\, g^{\beta_L}$ to $g \propto r^{1/(1-\beta_L)}$, hence

$$C \propto r^{\alpha}, \qquad \alpha = \frac{1}{1-\beta_L} \tag{1}$$

The corrective framework, when it leaves wall-clock time, passes to the depth clock $\tau = \ln(C/C_0)$. Substituting:

$$\tau = \alpha \ln r + \text{const} \tag{2}$$

□

The corrective framework's depth clock is the logarithm of the scaling framework's recursion depth, scaled by α. The two frameworks were already in one variable, up to a log, and neither noticed.

Remark 1 (Premise note (20 September 2026)). A review of the two results this paper labels exact finds that Result 1 is exact in the form its premise states, and this note states that premise rather than leaving it to be inferred. Integrated from a finite starting point rather than from an origin, $dg/dr = a\, g^{\beta_L}$ gives $C(r) = [C_0^{1-\beta_L} + a(1-\beta_L)(r-r_0)]^{1/(1-\beta_L)}$, which is $C = K(r+s_C)^{\alpha}$ carrying a shift s_C fixed by the

initial condition, and the clock relation then reads $\tau = \alpha \ln((r + s_C)/(r_0 + s_C))$. The unshifted power above is the aligned case $s_C = 0$, or the asymptotic case in which r is large beside s_C, and Result 1 is stated for those. The condition the two accumulations of Section 6.5 need is stated beside Result 2 there. Unchanged: Result 3, which follows directly from $dC/dr = aC^{\beta_L}$ and $dt/dr = bC^{-\delta}$, uses no pure-power premise, and is not withdrawn because Result 1 carries this shift.

Consequence. The control quantity $q = A/r_{\text{rate}} \propto C^{\beta_X - k}$ becomes $q \propto r^{\alpha(\beta_X - k)}$ in the aligned or asymptotic depth form. For fixed $\beta_X - k$ and positive α, changing coordinates preserves its sign. This is distinct from changing a system's growth parameters: at fixed δ, changing α changes $k = \delta - 1/\alpha$ and may change the criterion. Paper X's speed-prefactor result instead varies b at fixed exponents in its stated drift domain. Neither fact makes α a mere speed coefficient.

6 The correction exponent is a ratio of leverage complements

6.1 Not an assumption: a scope condition, and the reason matters

An earlier draft stated this as "Assumption A1: correction is itself a recursive process", and that framing does not survive contact with the source. Read at source, the corrective framework introduces correction as a **constitutive law inside its master system**:

$$\dot{D} = \gamma_1 \dot{C} + \gamma_2 C + \gamma_3 \frac{\dot{C}}{C} D - AD, \qquad A = A_0 C^{\beta}, \qquad \dot{C} = bC^{1+k} \tag{3}$$

A is a specified function of C. It is not a state variable. There is no $\dot{A}$, no dA/dr, and no corrector accumulation anywhere in the framework; searched exhaustively on 8 August 2026 and found absent in every form.

So β_X is a free parameter of that model, and there is nothing in it to derive. Stating A1 as an assumption *inside* the framework would introduce β'_L, a quantity with no referent, and the join would not be wrong so much as vacuous. **That is what a hostile reader finds in ten minutes, and it is better found here.**

The correct move is to state it as a scope condition on the corrector, not an assumption about the model. A scope condition cannot be falsified by finding a system it does not cover.

6.2 The corrector taxonomy: a decision tree over what A may depend on

An earlier draft used two binary questions and it does not close the taxonomy. Its first question was *"does corrective strength depend on its own history?"* **Class 1 answers no. Class 2 also answers no**, because a provisioned checker's strength depends on capability, not on its own past. **So the two-binary scheme put classes 1 and 2 in the same cell while their β_X differ as 0 from free, and that difference is Result 2a.** A partition that loses the strongest result's own distinction cannot carry an elimination argument. Withdrawn.

The decision tree distinguishes the mechanisms modelled here by what A is permitted to depend on: nothing, capability, or accumulated correction history, with the last split by substrate. It is a modelling partition, not an exhaustive classification of control systems. Joint dependence, adaptive provisioning, preserved goals under self-modification and hybrid oversight need their own models; this tree does not eliminate them.

Dependence of correction strength, physical location and mutability are separate questions. A static mechanism illustrates class 1, provisioned checking class 2, and correction accumulating under S1 class 3. Section 10 uses location and mutability to organise design questions, without claiming an exact correspondence, a historically empty cell or an exhaustive elimination of alternatives.

Table 2. A decision tree over the modelled dependencies of correction strength A, with each class's correction exponent β_X.

A depends on	class	β_X
nothing	**1 · constant strength.** $A = A_0$	**0 by definition**; implementation type does not fix effective strength
capability only	**2 · capability-carried.** A larger reviewing model does the checking	**free primitive.** Chosen by provisioning, at the cost of a regress
its own history, same substrate	**3 · self-accumulating**	**related to leverage** by Result 2, given S1's form condition
its own history, other substrate	**4 · cross-class**	not bounded by the conjecture stated for class 3; no value is inferred

The distinction matters for a class audit. A question about dependence on the corrector's own history alone cannot separate a static checker from one provisioned with capability. The audit needs to record that dependency separately from location, mutability, external grounding and failure-mode dependence.

6.3 Result 2a: constant strength and relative misalignment

For a class 1 corrector as defined here, $\beta_X = 0$ exactly, because a strength depending on nothing is $A_0 C^0$. Reading an exponent off a constant, not an estimate.

Substituting into the corrective framework's published criterion $\beta_X > k$:

$$0 > k \iff k < 0 \tag{4}$$

and $\dot{C} = bC^{1+k}$ with $k < 0$ is sub-exponential. Under the standing gain-drift assumptions of Section 4:

Result 2a. *Under the standing gain-drift assumptions, a corrector with constant strength $A = A_0 > 0$ gives $D/C \to 0$ exactly when $k < 0$. At exponential growth ($k = 0$) and above, the limiting fraction is positive. This is a relative criterion, not a conclusion about harm.*

That is the conjecture-free form. One definition, one published theorem, and no clock relation. **The $\delta < 1/\alpha$ version is its corollary**, obtained through Result 3 and therefore carrying one further derivation. **An earlier draft published the corollary as the free result, which overstated what the free result rests on.**

6.4 Fixed rules do not determine effective scaling

A rule or model can stay fixed while its outputs, memory, search, verification effort or surrounding resources change. Its effective correction strength can therefore vary with capability. A rule being fixed does not by itself establish $\beta_X = 0$, still less $\beta_X < 0$. The taxonomy classifies the effective law $A(C)$, which must be specified or measured.

If a mechanism instead has decreasing power-law strength, $\beta_X < 0$, the standing relative criterion is stricter than for constant strength:

$$k < \beta_X \leq 0 \tag{5}$$

The displayed weak inequality includes the constant-strength case; decrease requires strict negativity. The correlations between ethics and capability reported in Paper VI do not identify $A(C)$ or its exponent without a demonstrated measurement mapping (Eastwood 2026e). No negative correction exponent

is inferred from them here. Result 2a applies to constant effective strength, not to every filter, written constitution, guardrail or fixed-weight model.

Class 2 requires an account of how its checker is provisioned and remains effective. That requirement can move the correction problem to another system; it is not a proof that every externally provisioned scheme fails. The scalable-oversight comparison in Section 2.2 and the design questions in Section 10 remain relevant.

6.5 The scope condition, stated so it cannot be mis-refuted

Scope condition S1. Results 2, 5, 6 and 7 apply to **class 3 correctors only**: those whose correction strength accumulates from their own prior corrections, on the same substrate as the capability they correct.

Finding a class 1 or class 2 system does not refute them. It places the system outside their scope. And β'_L is undefined for class 1, so the join is inapplicable there rather than false.

The engineering contribution is conditional. Correction embedded inside a recursive loop is represented by class 3 only where its strength actually accumulates from its history on the same substrate and satisfies S1's form condition. An entangled objective is a candidate implementation, not proof of those premises: writing $C \times S$ does not by itself establish accumulated correction or preservation of the objective under modification. Result 2 relates the exponents where the premises hold.

That was not arranged. The engineering programme and this algebra were developed separately and meet at the class boundary.

Result 2 (exact, given S1's form condition)**.** *Correction accumulates as* $A \propto r^{1/(1-\beta'_L)}$ *while capability accumulates as* $C \propto r^{1/(1-\beta_L)}$*. Eliminating* r*:*

$$\beta_X = \frac{1-\beta_L}{1-\beta'_L} \tag{6}$$

The correction exponent is the ratio of the two leverage complements.

The neutral case is a check the derivation could have failed and does not. When correction leverages accumulated context exactly as well as capability does, $\beta'_L = \beta_L$, the expression gives $\beta_X = 1$ exactly, not an arbitrary constant. Correction strength then grows linearly with capability, which is the natural reading of "keeping pace".

Remark 2 (Premise note (20 September 2026)). The elimination above treats both accumulations as pure powers of one r, which is the aligned or asymptotic case of the finite-window solution stated beside Result 1 in Section 5. Written with their shifts, $C = K(r + s_C)^{\alpha}$ and $A = L(r + s_A)^{\alpha'}$ with $\alpha' = 1/(1-\beta'_L)$, the local slope is $d\log A/d\log C = (\alpha'/\alpha)(r+s_C)/(r+s_A)$, which is the constant ratio β_X above only where the two shifts align or in the limit where r is large beside both; one change of origin cannot in general align two different shifts. S1's form condition is read as carrying that alignment or that limit. Unchanged: $\beta_X = (1-\beta_L)/(1-\beta'_L)$ under the stated premise, the neutral case $\beta'_L = \beta_L$ giving $\beta_X = 1$, and Result 3, which uses no pure-power premise.

7 The self-acceleration exponent

The two frameworks still cannot be compared, because k is defined against wall-clock time and every quantity above is clock-free. The missing object is the map from recursive depth to time, which neither framework specifies. **Recursive self-improvement is precisely a system improving its own capacity to improve, and shortening its own cycle is one of the things that improves.**

Definition 1 (Self-acceleration exponent)**.** Let each recursive step take duration

$$\Delta t \propto C^{-\delta}, \qquad \delta \geq 0 \tag{7}$$

measured on a fixed substrate. δ is the self-acceleration exponent: the rate at which a system compresses its own iteration time as it becomes more capable, **through its own outputs.** $\delta = 0$ is constant cost per step. $\delta > 0$ is self-acceleration.

Clock and measurement scope. In the continuous-depth model, Δt represents dt/dr; using durations of discrete cycles requires that approximation to be justified. A fixed multiplicative change of throughput changes a coefficient, while time-varying hardware or external optimisation can change the apparent exponent. Attributing compression to the system's own outputs requires a protocol that controls or records those changes. Frozen weights alone do not determine the sign: a fixed model may produce faster procedures or reusable artefacts. The definition names the nonnegative self-acceleration branch; a fitted negative slope parameter describes slowing cycles and falls outside that branch, although the clock identity extends algebraically to it. An empirical mismatch under a changing substrate does not refute that identity.

Result 3 (exact, given the definition)**.**

$$\dot{C} = \frac{dC/dr}{\Delta t} \propto C^{1-1/\alpha} \cdot C^{\delta} = C^{1+(\delta-1/\alpha)} \tag{8}$$

and matching against the corrective framework's $\dot{C} = bC^{1+k}$*:*

$$k = \delta - \frac{1}{\alpha} = \delta - (1 - \beta_L) \tag{9}$$

This is k in its own clock, expressed in scaling-framework quantities plus one new observable. It is a derivation and not a transported label.

Table 3. The growth regimes set by the self-acceleration exponent δ (Result 3).

δ	k	regime
0	$-1/\alpha < 0$	constant cost per step. Capability grows as a power law
$< 1/\alpha$	< 0	still power law. Correctability requires $\beta_X > k$
$= 1/\alpha$	0	exponential in time. **The threshold**
$> 1/\alpha$	> 0	super-exponential. Correction must out-scale it

Result 4. *The threshold is* $\delta = 1/\alpha = 1 - \beta_L$*.* ***At $\alpha = 2$ it is exactly 0.5: if cycle time falls faster than the square root of capability, growth ceases to be a power law.***

The counterintuitive consequence. The threshold is the *reciprocal* of the scaling exponent, so **a system that is more efficient at recursion has a lower tolerance for self-acceleration.** The two compound with each other rather than offsetting.

7.1 Negative leverage as a conditional mathematical extension

The model declared above uses $\beta_L \in [0, 1)$ and therefore $\alpha \geq 1$. A mathematical extension to negative leverage is possible without substituting an empirical error-decay estimate for capability growth. No measurement cited here establishes that extension in a real self-improving system.

The same differential equation admits $\beta_L < 0$ on positive capability. In the aligned-origin or asymptotic sense specified in Section 5,

$$\frac{dg}{dr} = a\, g^{\beta_L}, \quad \beta_L < 0 \implies g \propto r^{1/(1-\beta_L)}, \quad \alpha < 1 \tag{10}$$

The positive-state ODE remains well posed under this extension. No parameter value is imported from the error-decay fit discussed below. The numerical example in Remark 3 illustrates the extended model without estimating any system.

Negative leverage in this model means that the marginal increment dg/dr decreases as the accumulated quantity g rises. It does not show that a frozen model uses less context, changes its reasoning rules or improves its own architecture. The extension gives three mathematical regimes:

Table 4. Three conditional leverage regimes of the extended model, $\beta_L < 1$. No row is an empirical classification.

β_L	α	reading
< 0	< 1	sub-additive asymptotic growth; the marginal increment falls
$= 0$	1	additive growth; a constant marginal increment
$(0, 1)$	> 1	super-additive asymptotic growth; the marginal increment rises

7.2 What the measurement is a measurement of

Paper II estimates error decay against resources, not capability growth against recursive depth. Its model is $E(R) = E_0 R^{-\alpha_{\text{err}}}$, with E an error rate and R the resource measure, tokens or samples. The subscript here distinguishes that estimand from this paper's capability-growth exponent α.

For Gemini 3 Flash, the reported sequential regression estimate of α_{err} is 0.49. The weights were fixed, and the measurement concerns test-time resource use. A fixed-weight model reasoning for longer is not evidence that it is modifying its own weights or improvement procedure. More fundamentally, no mapping from that error-decay estimand to $C \propto r^{\alpha}$ is established here, even within the fixed-weight regime.

The earlier inference of a measured negative β_L is withdrawn. Neither $\beta_L = 1 - 1/0.49$ nor a self-acceleration threshold $1/0.49$ may be inferred from the reported regression. Establishing either would require a justified capability scale, the appropriate recursive-depth variable and a demonstrated bridge between the two models. The measurements remain evidence about their own error-decay estimand, with the limitations below.

7.3 The estimates, their precision and a labelled hypothetical transformation

On Gemini 3 Flash, Paper II reports an endpoint sequential error-decay estimate of 0.59 with a bootstrap 95 per cent interval of $[-1.3, 2.9]$, a sequential regression estimate of 0.49 with standard error 0.20 and coefficient of determination 0.86, and a parallel endpoint estimate of 0.31. The regression uses three nonzero-error levels. The bootstrap interval belongs to the endpoint estimate, not the regression estimate. The sequential and parallel arms were not matched in total compute.

Under the fitted error-decay form alone, doubling resources would give the suppression ratio $E(R)/E(2R) = 2^{\alpha_{\text{err}}}$. A value of 0.49 would therefore imply a hypothetical ratio of about 1.40. This is not a 1.40-fold capability gain or accuracy gain, and it is not an observed effect of doubling a self-improvement cycle. Transforming the endpoint interval gives the following hypothetical ratios, conditional on the same power-law form:

Table 5. Hypothetical error-suppression ratios under $E(R) = E_0 R^{-\alpha_{\text{err}}}$ at the ends of Paper II's bootstrap 95 per cent interval for the Gemini 3 Flash endpoint estimate. These are transformations of an uncertain estimate, not capability gains or measurements of self-improvement.

α_{err}	hypothetical $E(R)/E(2R)$
−1.3, interval floor	0.41: the model permits error to increase with resources
2.9, interval ceiling	7.46: the model permits much stronger error suppression

The interval is too broad to establish a precise suppression benefit. It spans degradation and substantial suppression under the assumed form. The retracted $\alpha \approx 2.24$ estimate is not reinstated: the programme's earlier rule that a wide interval may accommodate both a claim and its negation remains applicable. Neither transformed endpoint establishes a capability exponent, a correction exponent or a location relative to this paper's thresholds.

Of the five frontier models in the tier-2 capability arm of Paper II, a sixth model having entered only the alignment arm, four did not provide a usable regression exponent: ceiling saturation at 94 to 100 per cent, a binary step, scores of 100 per cent throughout, or no trend. The Gemini fit's endpoint interval remains $[-1.3, 2.9]$. This is a limitation of those measurements and their benchmarks, not a conclusion about every frontier model or the field's entire evidence.

The result motivates measurements with headroom and adequate precision on a declared estimand. It cannot decide whether capability growth in recursive self-improvement is sub-additive, additive or super-additive. A correction note of 3 October 2026 is kept whole in Appendix A.4; Appendix A.5 preserves the wording corrected in this edition.

7.4 Two distinct thresholds, with no required ordering

The thresholds depend on the capability-growth exponent. The boundary $\delta = 0.5$ belongs to $\alpha = 2$; it is not a measured threshold for the systems in the error-decay study. Conditional on $0 < \alpha < 1$, the model gives $1/\alpha > 1$, but no cited measurement locates a real system in that regime.

(1) β_L crosses zero. In the extended model, the asymptotic capability-depth exponent crosses one, from sub-additive to super-additive growth. Its measurement requires the capability-growth estimand.

(2) δ crosses $1/\alpha$. Calendar-time growth becomes super-exponential. This crossing is meaningful for every $\beta_L < 1$, whether leverage is negative, zero or positive.

Remark 3 (Ordering, corrected 1 November 2026). The threshold $k = 0$ is $\beta_L + \delta = 1$ for every $\beta_L < 1$. Positive leverage is not a prerequisite. Taking the hypothetical values $\beta_L = -1$, $\alpha = 1/(1 - (-1)) = 1/2$ and $\delta = 5/2$ gives $k = 5/2 - 2 = 1/2 > 0$ and growth proportional to $C^{3/2}$. These equation values show the point without estimating a system. The ordering note of 19 September 2026 already supplied this counterexample; its complete historical wording is retained in Appendix A.5. Its additional claim about the ordering in measured systems is not retained as a current empirical conclusion, because the cited error-decay evidence does not measure β_L.

The model identifies two quantities to measure, not two alarms that must occur in sequence. Their empirical ordering remains open. The $\delta = 0.5$ example remains an illustration at $\alpha = 2$, a candidate value rather than an observed capability-growth exponent here.

8 The two boundaries, and the gap between them

Remark 4 (Interpretation rule, added 8 August 2026). Under the standing correction assumptions of Section 4, $k = 0$ is a growth-regime boundary and $\beta_X = k$ is the boundary for vanishing relative misalignment. Crossing the former changes the model's growth law and time available to react; it does not settle the relative criterion. For $\beta_X > k$, $D/C \to 0$ on either side, while absolute D may still grow. Neither boundary is a general safety threshold. The earlier corridor and safety wording is retained as history in Appendix A.5.

Result 5 (given S1's form condition)**.** *Combining Results 2 and 3 with the corrective criterion $\beta_X > k$, and writing $u = 1 - \beta_L$, $u' = 1 - \beta'_L$:* ***within the contested regime $\delta > 1/\alpha$ (equivalently $k > 0$), which is a scope restriction and not a consequence of the criterion****, the criterion $\beta_X > k$ is equivalent to $\delta < 1/\alpha + \beta_X$. Scope and criterion together give the window:*

$$\frac{1}{\alpha} < \delta < \frac{1}{\alpha} + \beta_X \tag{11}$$

The lower edge changes the growth regime; the upper edge ends vanishing relative misalignment. Between them the standing model has super-exponential growth and $D/C \to 0$, with no general conclusion about absolute burden or harm.

Result 6. *The gap between the two boundaries is*

$$\left(\frac{1}{\alpha} + \beta_X\right) - \frac{1}{\alpha} = \beta_X \tag{12}$$

The width of the model window with $k > 0$ and vanishing relative misalignment is exactly the correction exponent.

In the neutral case $\beta'_L = \beta_L$ it is exactly 1, independent of β_L, so the boundaries' *position* moves with the scaling exponent while the *gap* between them does not.

This is a conditional parameter comparison. Holding the growth parameters fixed, increasing β_X widens the relative-correction window one-for-one. A value of 1 gives width 1; a value of 2 gives width 2 if S1 permits it. The algebra does not establish that an engineering change can increase correction without slowing capability growth or changing other parameters. Takeoff speed alone does not determine the relative outcome, and a low relative error does not establish safety.

8.1 Result 7: a conditional bound on the gap

Result 7. ***The substrate-ceiling conjecture and the width identity combine, and neither gives this result alone.*** *The ceiling, stated between the two leverage fractions as $\beta'_L \le \beta_L$, gives $1 - \beta'_L \ge 1 - \beta_L$ and therefore by Result 2*

$$\beta_X \le 1 \tag{13}$$

with equality exactly when $\beta'_L = \beta_L$. *Within S1 and this additional ceiling conjecture:*

> ***Correction scales at most linearly with capability in this conditional family. The conclusion does not cover every internal corrector.***

Combining with Result 6, whose width is β_X:

$$\textit{boundary gap} = \beta_X \le 1 \tag{14}$$

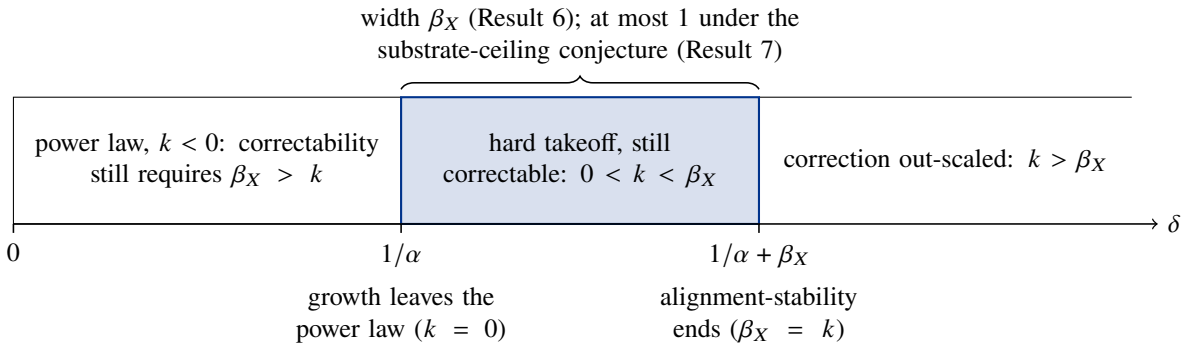


Figure 1. The two model boundaries for a class 3 corrector satisfying S1, positive β_X, and the standing gain-drift assumptions of Section 4. Below $\delta = 1/\alpha$, growth is a power law; at equality it is exponential. Above it growth is super-exponential. The fraction D/C tends to zero exactly below the upper boundary $1/\alpha + \beta_X$, including below the growth boundary; absolute D may still grow. The hard-takeoff window has width β_X (Result 6), at most one only under the additional conjecture $\beta'_L \leq \beta_L$ (Result 7). The figure gives no harm or safety verdict. Schematic: no value is a measurement.

Table 6. The growth boundary and the upper envelope for the relative-correction boundary, conditional on S1 and the substrate-ceiling conjecture. The second column enters super-exponential growth; the last excludes $D/C \to 0$ in that family, not every internal corrector. The actual upper edge can lie below the envelope.

β_L	super-exponential regime, $k > 0$	at or above conjectural envelope
0.25	$\delta > 0.75$	$\delta \geq 1.75$
0.50	$\boldsymbol{\delta > 0.50}$	$\boldsymbol{\delta \geq 1.50}$
0.75	$\delta > 0.25$	$\delta \geq 1.25$

Under S1 and the ceiling conjecture, the relative-correction window is at most one unit of δ wide. Its actual upper edge is $1 - \beta_L + \beta_X$, no greater than the envelope $2 - \beta_L$; it reaches that envelope only when $\beta'_L = \beta_L$.

The growth boundary and the conjectural upper envelope are exactly one apart. The actual window has width β_X and reaches one only in the neutral case $\beta'_L = \beta_L$.

Within S1 and the ceiling conjecture, $k < 1$ is necessary for vanishing relative misalignment. It is not sufficient when $\beta_X < 1$; the actual condition remains $k < \beta_X$. The envelope $k = 1$, corresponding to $\dot{C} \propto C^2$, is a conditional model bound, not a universal limit on correctors.

Status. Result 7 is conditional on the additional substrate-ceiling conjecture $\beta'_L \leq \beta_L$ for the class 3 model. Neither shared substrate nor independence of error channels derives that inequality. It is distinct from the registered P6 conjecture on the target-axis correction-leverage exponent, discussed in Section 12.1; no identification of those exponents is assumed. **Results 5 and 6 stand without this additional ceiling.** The deciding measurements remain draft registrations awaiting human submission.

8.2 The identity's status: PREREGISTERED BY TIMESTAMP FOR TESTING, not an established fact

The programme's stability ceiling and its conversion exponent meet at the same number, and the temptation on every surface is to say that one constant does two jobs. The claim's status is stricter than that and must be stated strictly: the identity is PREREGISTERED BY TIMESTAMP FOR TESTING, in advance of the data, with outcomes that can kill it. Two drafted registrations carried it, and one, study AG, carries it now. Study AE, since withdrawn from the programme, set out the reciprocity requirement between the

separately measured leverage cap and stability boundary, on the log scale, with three outcomes per cell (supported, refuted, insufficient precision) and an aggregation rule under which one refuted cell refutes the identity; its independence structure was disclosed in advance, including enumeration of any endpoint overlap between the two feeder designs, so that common-method variance would be assessed before analysis rather than raised afterwards as the explanation for agreement. Study AG (Unit 16) measures the leverage fraction and the correction-leverage exponent on one and the same system, each with its own instrument, and tests whether the growth exponent the leverage fraction sustains respects the ceiling the correction-leverage exponent sets; the two derivations share one functional form, so what it tests is their ordering, and it is designed to find a violation.

Surfaces therefore say: the identity is a claim preregistered by timestamp for testing, the deciding measurements are drafted and dated, and if the channels disagree the unification is withdrawn from every surface it has reached, with the same prominence as the framing. Surfaces never say the identity is established, and never present the agreement of an asserted one half with the regression estimate of 0.49 (Section 7.3) as evidence, because the 0.49 estimate concerns error decay, while the asserted ceiling concerns correction capacity; no bridge between those estimands is established.

In this paper's terms. The clock relation $k = \delta - 1/\alpha$ (Result 3), the ratio $\beta_X = (1 - \beta_L)/(1 - \beta'_L)$ (Result 2) and the width identity boundary gap $= \beta_X$ (Result 6) are algebra given the definitions of Sections 6 and 7, and their status is stated as algebra in Section 14.1. The identity between the programme's separately derived critical exponent and this paper's threshold $1/\alpha$ is governed by the paragraph above: study AG is its test, it is not asserted as established here, and the deciding measurements are prepared as draft registrations awaiting human submission. For the same reason, any agreement between the asserted $1/2$ and the error-decay regression estimate $\alpha_{\text{err}} \approx 0.49$, whose row is printed whole in Section 7.3, is not put forward here as corroboration.

9 What decides the outcome is a ratio, not a level

The strict asymptotic criterion compares exponents. In the standing gain-only model, the sign of $\beta_X - k$ determines whether D/C vanishes. Coefficients still determine transients and the positive floor at equality; absolute burden is a different outcome. The equality and extended-drift cases are specified in F5 and Section 12.

This model criterion contains no absolute capability threshold. That does not prove that the specified correction law can be implemented at every physical scale, or that either a slow or a fast takeoff is safe. Those claims require evidence beyond the equations.

9.1 Result 8: throughput multipliers and self-acceleration caps

Both this paper and its sources have stated that speed-independence means rate interventions cannot change the verdict. That is half the statement, and it is the unhelpful half.

By Result 3, $k = \delta - 1/\alpha$. **δ is inside k.** So the verdict $\text{sign}(\beta_X - k)$ is not indifferent to every rate-like quantity. A constant multiplier leaves it unchanged; a change to the cycle-time exponent need not.

Result 8. ***A constant throughput multiplier leaves the exponent criterion unchanged. A cap on self-acceleration can change it.*** *This comparison holds within the constant-exponent model; it does not rule out other interventions, including the additive serial latency analysed below.*

A nonshrinking cycle is a sufficient model condition for sub-exponential growth. From $k = \delta - 1/\alpha$: **enforce $\delta \le 0$, that is, a self-improvement loop's cycle time may never shrink, and $k \le -1/\alpha < 0$ for every $\alpha > 0$.** Within the model and its fixed-substrate scope, preventing cycle compression guarantees

Table 7. Two rate-like interventions and their effect on the verdict $\mathrm{sign}(\beta_X - k)$ (Result 8).

intervention	acts on	effect on the verdict
multiply throughput by a constant, with the exponents fixed	the coefficient b in $\dot{C} = bC^{1+k}$	**none.** b does not appear in the verdict
cap self-acceleration: bound how fast a system may shorten its own cycle	**δ, inside k**	**direct.** Reducing δ reduces k one-for-one

sub-exponential growth for $\alpha > 0$. It does not by itself guarantee correctability, which still requires $\beta_X > k$. Operational enforcement requires an identified cycle, control over its scheduling and verification that the enforced delay cannot be bypassed. Those implementation properties are not proved by the exponent relation.

A calibrated version exists for operators willing to measure: a minimum cycle duration shrinking no faster than $C^{-1/\alpha}$ permits self-acceleration up to the threshold. **The calibrated cap requires an identified α and its uncertainty.** This paper supplies neither a calibration nor evidence that a regulator cannot obtain one.

Its usefulness side. The nonshrinking-cycle cap permits continued capability growth, as a power law in the substrate-fixed constant-exponent model. The calibrated cap also permits exponential growth at equality, $\delta = 1/\alpha$. A fixed multiplicative throughput improvement changes only a coefficient; a time-varying intervention can change the apparent exponent (Section 7). The nonshrinking cap excludes compression of the loop's own cycle. This paper does not measure either cap's cost to an operator.

9.2 Verification overhead and a conditional reading of earlier simulations

A proposed retrodiction, with its identification condition still open. The simulation result was known before this derivation existed.

The honey-architecture simulations (Paper VI) compared a baseline, an entangled objective, and an entangled objective **plus a fixed verification tax**. The complexity-scaling arm found the advantage constant rather than compounding, and the programme published that against its own interest.

A proportional verification overhead multiplies cycle duration and leaves its exponent unchanged. An additive serial latency does not. The latter changes the local self-acceleration exponent and removes the model's finite-time divergence, as the calculation below shows. The honey-architecture result is not an established example of either case: this paper has not identified its simulated tax as per-cycle wall-clock latency rather than a penalty in a combined score.

Remark 5 (Scope note (19 September 2026)). The distinction recorded on 19 September 2026 is applied to the current argument here. Write the taxed step duration as $\Delta t_h = bC^{-\delta} + h$, with $h > 0$ a serial latency that cannot be parallelised away. The local self-acceleration exponent, defined as $-d \log \Delta t_h / d \log C$, is then $\delta b C^{-\delta}/(bC^{-\delta} + h)$, which decays to zero as C grows instead of holding at δ. With $dr = dC/(aC^{\beta_L})$, the elapsed time to capability C acquires the further term $(h/a)\int u^{-\beta_L}\,du$, whose antiderivative $u^{1-\beta_L}/(1-\beta_L)$ diverges at the upper limit for every $\beta_L < 1$; so within the bound $\beta_L \in [0, 1)$ declared in Section 4, a positive additive latency removes the finite-time divergence in every region where the untaxed model has one, including the whole of the super-exponential regime $k > 0$. A proportional overhead $\Delta t_q = q \cdot bC^{-\delta}$ leaves δ untouched and changes only b, so a proportional tax changes only the coefficient; so is the additive case at $\delta = 0$, where b simply becomes $b + h$; and over a bounded range with h much smaller than Δt a coefficient-only reading may still be defensible if the range and the error are quantified. The earlier wording used "fixed" without this qualification; Appendix A.5 preserves that wording. F4 in Section 12 and the paragraph in Section 15.1 stating that iteration time may

not follow a constant-exponent law both record that Result 3 is algebra given a single constant exponent, and an additive latency is precisely a departure from that parameterisation. Unchanged: Result 3 itself, the clock relation of Section 5, and the Section 9.1 argument that enforcing $\delta \leq 0$ buys the sub-exponential regime unconditionally, which the additive result strengthens, since an enforced positive serial latency is a second and constructive route to the same guarantee. The honey-architecture paragraph above is a separate matter: this paper does not establish that the honey-architecture tax is an additive per-cycle wall-clock constant rather than a penalty applied to a combined score, and the retrodiction stands pending that identification.

The model suggests a comparison of proportional overhead with irreducible serial latency and capability-dependent verification cost. Each needs a declared mapping from the implementation to cycle duration; a cost assigned to a score is not automatically a time cost. The programme has described such a comparison as a separate draft registration awaiting human submission. No result of that comparison, or claim that no other programme has run one, is made here.

The engineering, policy and algebraic questions meet at the cycle-time model. Whether verification changes a coefficient, an exponent or the form of the clock depends on its implementation. That mapping is a measurement obligation, not an identity supplied by the word "cost".

10 The governor as a scoped modelling contribution

The mechanism considered here is a **governor**: a restraining mechanism whose effect changes with the output it restrains. Watt's centrifugal governor is the familiar mechanical illustration, with flyballs responding to engine speed and adjusting the throttle. The comparison motivates a question about recursive systems: can correction remain effective while both the governed system and the corrector change?

Maxwell (1868, pp. 270 to 272) studies regulation of machine speed under changing drive or resistance. His stability test requires disturbances to decay; the characteristic roots' real parts decide whether motions decay or grow, including oscillatory motions. He also describes how stronger governing action can produce growing oscillations. Wiener (1948, Chapter IV, pp. 95 to 98 of the second edition) treats feedback of actual performance, including Watt's governor, and the possibility that badly adjusted feedback produces oscillation. These comparisons concern the dynamics of deviation from a controlled quantity. Here $\beta_X - k$ compares the scaling of correction strength with the specific capability-growth rate, within the stated drift model, and decides whether the relative misalignment D/C vanishes. It is not a characteristic root, and that relative conclusion does not by itself bound the absolute burden D (Section 4).

10.1 Location and mutability organise questions, not a historical gap

Table 8. Two design axes for a governor. The cells organise this paper's modelling questions; none is claimed to be historically empty or generally impossible.

	fixed over the modelled window	mutable by the governed system
external to the system	What provisioning keeps correction effective as capability changes?	What access permits modification, and what constraints survive it?
part of the system	What preserves the mechanism or its objective during self-modification?	Under what conditions does corrective strength accumulate with capability?

Location, access and mutability do not collapse into one axis. An externally located regulator may expose a modifiable interface; a self-modifying system need not modify every component it can reach. The

dependency classes of Section 6.2 therefore do not map one-to-one onto this table. Adaptive provisioning, hybrid arrangements and preservation of a fixed objective remain alternatives that require explicit analysis.

Earlier work already addresses changing regulation and self-modification, with different objects and guarantees. Conant and Ashby (1970, Section 4) explicitly allow a time-varying model when the system-event distribution changes slowly enough to be treated as constant over each relevant period. That observation concerns the regulator's mapping and the outcome entropy; it does not fix a correction-strength exponent or a recursive cycle time.

Yudkowsky and Herreshoff (2013, Sections 2, 3 and 7) study agents that approve similarly designed successors with shared goals. Its initial logical setting is fully known and deterministic; the report identifies the Löbian obstacle to a straightforward self-trust argument, gives technical ways around it, and states the limitations of its probabilistic extension. It is a restricted formal treatment of goal-preserving succession, not a measured correction-scaling result. Everitt et al. (2016, Assumption 8 and Theorem 16) treat policies and utilities that can be modified. With modification-independent belief and initial utility, and an initially optimal policy for the realistic value function, the agent anticipates modifications and evaluates their consequences using its current utility. The theorem preserves initial-utility optimality along the histories that the evolving policy produces. It permits value-equivalent changes and gives no guarantee that the initial utility captures human values; off-policy behaviour lies outside that conclusion.

Preserving a chosen objective is also distinct from accepting correction by people. Soares et al. (2015, Sections 1.1, 2 and 4 to 6) state corrigibility desiderata that include accepting shutdown, avoiding incentives to cause or prevent it, and passing shutdown behaviour to successors. Its examples expose failures of candidate utility combinations, including utility indifference, rather than establish a general solution. Orseau and Armstrong (2016, Assumption 9 and Theorems 14 and 17) prove asymptotic safe interruptibility for Q-learning and a modified Sarsa rule under finite communicating Markov decision processes, bounded rewards, the stated learning-rate conditions, a base policy greedy in the limit, and an interruption-probability bound tending to one while every state-action pair remains infinitely explored. The result concerns learning the non-interrupted task policy while an interruption mechanism remains available; it does not guarantee objective preservation under arbitrary code changes, or alignment after that mechanism is lost.

This bounded comparison, checked against these primary texts on 1 November 2026, supplies neither an exhaustive history nor a priority verdict for the joined exponent relation. Its positive distinction is the question asked: how correction strength and capability growth scale together under the clock and S1 premises. Goal preservation, corrigibility and interruptibility impose additional requirements that the exponent comparison alone does not settle. The scalable-oversight comparison in Section 2.2 supplies a further, instrument-specific comparison.

10.2 An embedded objective as a candidate, without elimination of alternatives

1. External correction may become inadequate if its effective strength fails to keep pace. That is a conditional statement about scaling, not a general impossibility result for external oversight.
2. A governor exposed to modification may be weakened, preserved or improved. Reachability alone decides none of these outcomes; the permitted changes and the objective governing them must be modelled.
3. An objective of the form $C \times S$ is one candidate for making the measured safety term consequential to optimisation. At fixed positive C, reducing S reduces that product. If C also changes, the product can increase while S falls, and the objective's own preservation is a further premise.

The contribution is the conditional exponent comparison applied to this design question. Where an implementation satisfies S1, Result 2 relates correction to capability, and Results 5 and 6 state the

correctable window in the model. An entangled objective is one candidate implementation; no elimination argument proves it uniquely necessary or proves that its removal must degrade capability. Whether an implementation occupies class 3 and preserves its correction remains to be measured.

People keep oversight, containment and shutdown for as long as they physically can. Embedded correction is the contingency for control lost against our will (Section 1), and this modelling argument supplies no reason to relinquish those controls.

11 Measuring the self-acceleration exponent

δ is the negative slope of cycle duration against capability on logarithmic axes. It is a model parameter with an operational interpretation, not an already validated instrument.

A measurement needs an admissible self-improvement loop; a fixed or explicitly accounted substrate; a definition of each cycle and its stopping rule; paired durations and a positive capability measure on an identified scale; and an evaluation protocol protected against training leakage, changing difficulty and score saturation. A held-out battery alone does not establish a ratio scale for capability. The chosen scale and sensitivity to admissible alternatives must be declared before measurement. Regressing $\log \Delta t$ on $\log C$ estimates the negative slope only where the functional form is adequate; uncertainty must account for error in both variables, dependence across cycles and departures from a constant exponent. Attributing the slope to endogenous improvement additionally requires separating external changes in resources or procedures.

Existing logs may be sufficient, but that is a condition to check. Recording time needs no special clock. Identifying capability, the cycle and effective correction may require a new benchmark, instrument or evaluation protocol. No census of available records is supplied here, so this paper claims neither that suitable pairs are absent everywhere nor that every loop already records them. Suitable records could also identify α or β_X; δ has no established exclusivity as the only accessible parameter.

Frozen weights do not predict a nonpositive exponent. A fixed model or rule can generate faster algorithms, reusable code or changing external memory, so effective cycle time may fall without a weight update. Conversely, a self-modifying loop may slow down. Neither sign is forecast here. Release-to-release intervals are not automatically the cycle times this model requires: changing resources, evaluation, selection and release policy can confound them. No shortening trend or typical number of releases is established in this paper.

Possible interpretations, conditional on an admissible measurement:

- An interval wholly at or below zero finds no positive compression exponent in the measured window under its uncertainty model. A strictly negative value describes slowing cycles, outside Definition 1's nonnegative branch; the clock relation still extends algebraically. It is not evidence that self-acceleration is impossible in other systems or later windows.
- A positive exponent below $1/\alpha$ places the fitted model in its power-law regime. Both parameters and their joint uncertainty must be identified before that ordering is asserted.
- Equality at $1/\alpha$ is the exponential model boundary, and an exponent above it is in the super-exponential model regime. A finite interval cannot establish exact equality. Registered equivalence or separation rules are needed for empirical decisions near the boundary.
- An interval spanning zero or a relevant boundary, insufficient precision, or a failed scale or model check leaves the corresponding classification inconclusive. No point estimate or uninformative interval is converted into a verdict.

These are interpretations, not a replacement for a registration's frozen admission, estimation and verdict rules. The relative-correction comparison needs β_X in every growth regime, together with its uncertainty

and the standing drift assumptions. Absolute burden and harm require their own outcomes.

Measuring δ can locate a system relative to the model's growth boundary once α is identified. No algebra forces δ to rise before a correction failure, since β_X and other model conditions may also change. The paired observable is not a validated warning system by itself.

12 Falsification criteria

Two different objects can fail. A mathematical result is refuted by a counterexample satisfying every stated premise or by an error in its derivation. An empirical application is rejected when its specified observable consequences fail under a valid, adequately precise protocol. A finite record cannot establish an infinite-capability limit. Before a measurement supplies a verdict, its registration must fix admission, estimands, observation range, uncertainty, rejection and inconclusive rules; no new numerical decision rule is introduced here.

F1. A corrector that does not satisfy S1's accumulation and form premises is outside the correction-ratio join. Results 2, 5, 6 and 7, as joined through S1, are not predictions for it. The underlying clock identity and the separately specified corrective model can still be assessed on their own premises.

F2. An admissible S1 system with $\beta'_L > \beta_L$ would contradict the additional substrate-ceiling conjecture for that class. Empirical rejection requires a registered comparison on the same depth, identified scales and comparable regimes, with uncertainty resolving the inequality. Shared physical substrate alone is insufficient admission. This does not refute the width identity without that ceiling.

F3. A mathematical solution with the standing gain-drift assumptions, $\delta > 1/\alpha + \beta_X$ and $D/C \to 0$ would refute the claimed asymptotic boundary. A fraction that falls over a finite observation window does not do so: a decreasing transient can approach a positive floor. An empirical test must instead specify a finite-window prediction of the fully parameterised model, including initial conditions and coefficients, and reject that application if its registered criterion is met. Neither falling relative error nor bounded relative error establishes vanishing absolute burden.

F4. For fixed exponents with $dC/dr = aC^{1-1/\alpha}$ and $dt/dr = \ell C^{-\delta}$, $a, \ell > 0$, the chain rule gives Result 3. A counterexample preserving those premises would refute the derivation. Observed faster-than-exponential growth together with an estimated $\delta < 1/\alpha$ is evidence against their joint empirical fit or measurement mapping, subject to the registered uncertainty rule; it does not make the algebra false. Changing substrate, a varying exponent or a failed continuous-depth approximation breaks a premise.

F5. The strict-domain speed claim concerns the vanishing versus nonvanishing fraction in the constant-exponent, gain-only model: $\gamma_1 > 0$, $\gamma_2 = \gamma_3 = 0$, $A_0, b > 0$, $\beta_X \neq k$, nonnegative finite initial D and $C \to \infty$. Replacing b by another positive constant while preserving all those premises cannot change the sign of $\beta_X - k$. A mathematical example that did change that verdict would refute the claim. Changed transients, fixed-capability error, or the positive floor at equality $\beta_X = k$, $\gamma_1 b/(A_0 + b)$, do not. A capacity cap preventing $C \to \infty$, a latency changing the exponent, or coefficient-dependent compounding at equality is a different case, not a counterexample to this strict-domain claim.

F6. For the same fixed α and β_X, the two boundaries defined by $k = 0$ and $k = \beta_X$ differ by β_X. A mathematical counterexample satisfying these definitions would refute Result 6. An empirically different gap rejects the empirical model application or common-parameter identification under its registered rule; it cannot refute subtraction. Varying parameters between systems is not a measurement of this model's fixed-parameter gap.

Extensions do not share every boundary. The earlier assertion that a compounding coefficient above unity defeats $\beta_X > k$ was incorrect. In Paper X's gain-drift model with $\gamma_2 = 0$, $\gamma_1 > 0$ and finite constant $\gamma_3 \geq 0$, $\beta_X > k$ still gives $D/C \to 0$. At equality, put $h = A_0/b + 1 - \gamma_3$: $h > 0$ gives the positive limit γ_1/h, $h = 0$ gives linear growth of d in τ, and $h < 0$ gives divergence. Below the exponent boundary, $\gamma_3 < 1$ gives a positive floor, while $\gamma_3 \geq 1$ gives divergence. Thus coefficients can change boundedness at equality even when exponents stay fixed. With positive level drift as well as positive gain drift, the vanishing-fraction condition is instead $\beta_X > \max(k, 0)$ in that constant-power model (Eastwood 2026f, Section 3 and Appendix A). The core results here use the narrower standing gain-only scope; extensions require their own stated conditions.

12.1 The eclipse: falsification numbers beside the prediction

The same-class ceiling is a conjecture, not a consequence of shared substrate alone. P6 of the theory registration, read at version 1.104, conjectures a correction-leverage exponent at or below one half for correctors admitted as same-class by its frozen taxonomy. Independence of comparable readings gives an uncertainty scaling in the number of readings. To obtain the registered capacity exponent, the argument additionally needs capacity to be inverse uncertainty, uncorrelated channels of comparable variance and quality, and channel count proportional to capability. The registration also requires the inverse-uncertainty and misalignment-removed elasticities to agree within a registered equivalence margin before one substitutes for the other. Shared substrate does not prove the needed error structure, and independently reducing uncertainty does not by itself establish a bound on correction service.

The class contrast and P6 have different verdict rules. The drafted contrast asks whether a cross-class correction exponent exceeds a same-class one. Where the ratio is identifiable, the null ratio is 1.00. An interval merely containing 1.00 does not refute an advantage. A practical-equivalence verdict requires the interval to lie wholly within a symmetric region fixed before data, with the unit of analysis, sensitivity demonstration and separate support, refutation and inconclusive rules specified in the deciding registration. A comparison too imprecise to distinguish the registered difference is inconclusive. No numerical margin is supplied by this paper, and P8's margin on an exponent is not transferred to a ratio. Preliminary evidence described in earlier versions as adverse remains precursor evidence, not a verdict under those rules.

P6's replicated refuter, exactly scoped. Under version 1.104, the same corrector's interval must lie wholly above one half on the first fault set and on a second set analysed alone: the same correction type, the same construction and settings, the same window and the first set's registered level. The second set is disjoint and sealed before any review. A first-set exceedance that does not replicate is inconclusive, with its first interval printed, and is never counted as "no counterexample found". The deciding cell must demonstrate in advance, on its own ladder, the detection probability for the replicated refutation as a whole, meaning that both intervals clear one half. A cell below that registered probability is examined and uninformative. At the feeder's stated precision, a point estimate above one half may still be insufficiently precise to refute; its interval is printed and it is never described as consistent with the bound merely for spanning one half.

Correction type, scale and admission remain binding. P6 and the ARC Bound take a separate word for each correction type. The counted route applies only where every term of the interval is a count of checkable departures from the frozen task specification, settled by frozen code, with removal requiring the corrected artefact still to pass its task verifiers. Missing, blocked, empty or unreadable outputs are missing, not removals. A disjoint registered pilot must establish that the count is proportional to the declared ratio-scale correction quantity, with its tolerance carried into the interval; the declared ceiling, transformed-scale sensitivity, capability scale, both-axis error and registered looks also govern admission. A result on checkable correction refutes P6 and the ARC Bound only for that type and domain, not for

judged correction where the safety claim lies. A judged term restores the scorer gate. The route opens no population: until all six axes of the same-class taxonomy are frozen and cited, P6 is not evaluable. These are necessary conditions summarised from the registration; its complete admission and verdict rules govern, and a significant exceedance alone is insufficient.

P6 admits no supported verdict from a finite null survey. Its outcomes are a valid replicated refutation, no counterexample found among the stated number of admissible correctors, or inconclusive for insufficient precision. The no-counterexample outcome is marked inconclusive on the scoring sheet. A class contrast can test an advantage within its scope; it cannot by itself establish the universal same-class cap.

Three exponents remain distinct. A residual-decay exponent describes how error changes with accumulated correction effort. The registration's target-axis correction-leverage exponent describes correction capacity against the corrected system's capability. This paper's β_X describes the constitutive correction strength $A(C)$ and, under S1's form condition, equals $(1-\beta_L)/(1-\beta'_L)$. Result 7's bound $\beta_X \leq 1$ additionally assumes $\beta'_L \leq \beta_L$; it is not derived from P6's one-half conjecture. Bounds on different estimands cannot be combined by taking the smaller number without a demonstrated conversion. Study AG addresses the separately specified relation on its own terms (Section 8.2); no equality is established here.

12.2 A programme hypothesis beyond the model

A separate programme hypothesis asks whether specified human-in-the-loop arrangements attain a higher correction exponent than specified model-only arrangements. Different physical substrates alone do not establish effective correction strength, independence of failure modes or the ordering of exponents. Testing the contrast requires an admitted class definition, comparable capability and correction scales, suitable paired observations and the registration's decision rules. This paper establishes neither the contrast nor the availability of a suitable existing dataset. The hypothesis is distinct from the conditional algebra above.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

13 What is withdrawn from earlier drafts

Narrowed, not withdrawn. 1. $k = \beta_L - 1$ was withdrawn in full in an earlier draft. **That was an over-correction.** With the step-to-time map made explicit it is the $\delta = 0$ case of Result 3 exactly, verified at $\alpha = 2$, 4 and 10. **It was under-specified rather than wrong, and it is restored here as the constant-cycle-time case with its assumption stated.** A withdrawal that discards a true special case loses work.

Withdrawn 1b. The conclusion drawn from it, that the power-law regime is "automatically alignment-stable", **is withdrawn and must not be restored.** It would be read as "systems in this regime are safe". **The defensible statement is narrower: a power law has no finite-depth singularity, so correction has unbounded depth in which to act, which is not misalignment tending to zero and says nothing about the correction being adequate.**

Withdrawn 2. The substrate ceiling stated as $\beta_X \leq \beta_L$. This compares an unbounded exponent to a bounded fraction. **Its absurd consequence, that at $\alpha = 2$ it forces a corrector with zero leverage, is what a type error looks like carried through.** The ceiling belongs between the two leverage fractions,

$\beta'_L \leq \beta_L$, and is prepared as a separate draft registration awaiting human submission as a conjecture in its own right.

Withdrawn 3. A formulation $k \leq \beta + \gamma \leq 1$ with $\alpha_{\text{crit}} = 1/\gamma$ was carried in working notes as though it were established. **It could not be located in either framework**, and the γ the corrective framework does define is a drift coefficient rather than an exponent. **It is not cited here and must not be reconstructed from a description of it.**

Recorded rather than deleted, because a withdrawal without the reasoning that produced it gets re-derived by the next reader.

14 What this paper does and does not show

14.1 A derivation, not a measurement

This paper derives the growth boundary $\delta = 1/\alpha$ within the specified clock model, and joins it to a relative-correction criterion under separate drift and S1 premises. Estimating the exponents requires the measurement conditions in Section 11. It is not a measurement claim: this paper supplies no paired measurement establishing the capability-growth, corrective and clock exponents in its central result. Section 12 states what would refute it, Section 13 records what earlier drafts withdrew, and Section 2 states what is not claimed as new.

Claims come in three sizes: results, laws, frames. This is a laws-and-frames paper: Results 1 and 3 are algebra; Results 2, 5 and 6 are algebra given scope condition S1's form condition; Result 2a needs nothing beyond the corrective framework itself and one definition; and Result 7 additionally rests on a conjecture that is flagged rather than folded in. Nothing here is measured, and the paper says so wherever the temptation to imply otherwise would arise. The governor discussion of Section 10.2 supplies a scoped model, without establishing a uniquely necessary architecture or a historically empty cell; the retrodiction of Section 9.2 stands pending identification of what the simulated tax models; and the identity between the programme's separately derived critical exponent and this paper's threshold is preregistered by timestamp for testing and is not asserted as established (Section 8.2).

Two of the five conditions set for this paper's deposit remain open and are stated here rather than waived: the deciding measurement is drafted, dated and prepared as a draft registration awaiting human submission, and no mathematician without a stake in the programme has yet read the algebra adversarially (Appendix B.2). The author took the disclosure decision to publish on 16 August 2026 with both open conditions known. The programme-wide correction record is at michaeldariuseastwood.com/research/corrections.

14.2 This paper and Laws IV and V

This paper bears on Law V, the ARC Embedding Law. It relates correction to capability where an embedded mechanism satisfies the accumulation and form premises of S1 (Section 6.5). Its governor model examines an embedded, mutable corrector as one candidate, without proving that its removal degrades capability or eliminating alternatives (Section 10.2). It also discusses a proposed retrodiction, conditional on identifying what the simulated tax represents, of a simulation result the programme had already published: in the complexity-scaling arm of the honey-architecture simulations of Paper VI, which varied the network's complexity and not recursive depth, the advantage of the entangled objective was constant rather than compounding (Section 9.2). It measures nothing. It reports no comparison of correction placed inside the loop with correction applied outside it as depth grows, which is P21, and it says nothing about build order, which is P12. It does not bear on Law IV, the ARC Persistence Law: its conditional analysis of constant or decreasing effective correction strength (Section 6.4) concerns the

specified $A(C)$, not whether gains installed by external post-training persist under later capability-directed training, which is P15. No result in this paper is a deciding test of P15, P12 or P21. The note of 3 October 2026 that first placed the paper against the two laws is kept in Appendix A.4.

14.3 The laws and the programme, as at 1 November 2026

What this programme names Laws are conjectures, and each is a law under test until a registered test returns its verdict. Laws I to III were registered on 8 September 2026 and Laws IV and V on 15 October 2026, in version 1.103 of the theory registration; its current version is 1.104, approved on 29 October 2026, which adds no prediction and changes no law's statement. Its terms are defined in a separate register, and the mathematics of Laws IV and V is in development (version 1.2, 29 October 2026). Law IV has not yet been tested. The counted test of P21, Unit 83, was registered on OSF on 22 October 2026 and ran on 27 and 28 October 2026: P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (results: DOI 10.17605/OSF.IO/98HKR). Two different axes are named here. The access axis is what Unit 83 varied: both reviewers are called each round, but the corrector sees the earlier kept entries with a marked new arrival and may revise it, while the monitor sees its accumulated register with the new arrival unmarked and can only accept or reject it. The performance axis is what it measured, the outcome, as against the mechanism axis, which was not evaluable. Law V, the ARC Embedding Law, remains a law under test: its placement limb, P21, is supported on the performance axis only, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism; its build-order limb, read on P12, is undecided, since P12 is untested. Unit 83's result concerns where correction is placed, and it decides none of this paper's results. The programme's first registered study, Unit 06, a classification audit that tests no law, was registered on 1 October 2026 and ran on 6 October 2026; it is reported in full as administratively incomplete, with no inferential verdict. Unit 82 was registered on OSF on 27 October 2026 and has not yet run. The measurement this paper defines, Unit 30, is a draft registration awaiting the author's submission; its design is preregistered by timestamp, first anchored on 12 August 2026 at block 962066, and it has not been run, so no result is claimed. Study AG (Unit 16) and the corrector-class contrast (Unit 01) are drafts preregistered by timestamp, first anchored on 12 August 2026 and on 28 August 2026. The mathematical quantities are defined, while their empirical identification remains conditional on the stated measurement requirements; established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

15 Limitations

15.1 The burden, stated in the direction that costs the author

The following are material limits on applying the results. They are not an exhaustive list of possible model failures:

1. **Effective correction does not satisfy S1.** A fixed procedure can produce accumulating corrections; its fixed implementation does not settle the question. If the specified accumulation and form premises fail, Result 2 is inapplicable to that system.
2. **Iteration time does not follow a constant-exponent law.** If Δt falls in a way not captured by a single δ, Result 3 holds only locally and the threshold is not a number but a moving target.
3. **The scaling framework's exponent is not what governs real recursive capability growth.** The reported error-decay regression estimate of 0.49 is a different estimand and cannot decide this

capability-growth claim (Section 7.3). **No paired measurement of the model's capability-growth and cycle-time exponents is supplied here.**

4. **The corrective framework's two-variable model is too coarse.** Its criterion may not survive a treatment with realistic state.
5. **δ may be unmeasurable in practice** if capability and cycle time cannot be separated, because a cycle that also raises capability confounds both axes.
6. **The hard-takeoff window may never be reached**, if no system sustains a sufficiently positive δ. Relative correction can still fail below it when $\beta_X \leq k$, so this does not make every corrective question disappear.

The burden remains with the model application. Its premises and measurement mapping need evidence; listing possible failures does not transfer that burden to a critic. A demonstrated derivation error or failed registered model test requires the corresponding correction or rejection, recorded on the programme's public corrections page.

15.2 Scope, stated in one place

The results hold inside the stated models and are no general guarantee. Results 2, 5, 6 and 7 apply to class 3 correctors only (scope condition S1); the core relative-correction conclusions use the standing gain-only drift assumptions of Section 4, with equality and extended drift treated separately in Section 12; every quantitative claim is in the substrate-fixed clock (Section 7); and Result 3 assumes a single constant exponent for iteration time, from which an additive serial latency departs (Section 9.2). The cited test-time estimates concern error decay and establish no value of the leverage fraction β_L (Section 7.2). Two original conditions set for the paper's deposit remain open (Section 14.1). The primary-source comparison in Section 10.1 establishes the scope of the cited results; it establishes neither historical priority for this model nor objective preservation, corrigibility or interruptibility for an implementation.

16 Conclusion

The clock relation $k = \delta - 1/\alpha$ connects two programme models. Under the stated constant-exponent clock, $\delta = 1/\alpha$ is the growth boundary. Under the separate gain-drift correction assumptions, $D/C \to 0$ exactly when $\beta_X > k$. With S1's form condition, the hard-takeoff window satisfying that relative criterion has width β_X; its conjectural bound of one needs the additional ceiling conjecture. These results do not establish bounded absolute misalignment, a measure of harm or a universal corrector limit. Their practical use is to identify quantities and assumptions a laboratory would need to test on an admissible recursive improvement loop. No paired measurement is supplied here, frozen weights predict neither sign of δ, and finite observations require a registered empirical model test rather than an inference about an unobserved limit.

Code and data. This paper is a derivation and has no code or data of its own. The figures it quotes are Paper II's estimator row for Gemini 3 Flash (Section 7.3) and the historical Paper VI adversarial-pressure range retained in Appendix A.5, each pinned in the deposit record (Appendix B.2). Section 6.4 explains why that range identifies no correction exponent here. The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/HT8WU and mirrored in the public repository github.com/MichaelDariusEastwood/arc-principle-validation, folder `papers/Paper-XIII-Self-Acceleration-Exponent`.

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Michael Darius Eastwood. Statement Paper · The ARC Theory · Five Laws of Recursive Growth, Correction and Persistence. OSF, 2026i. Cited edition: version 7.4, 31 October 2026. First published 14 August 2026. doi:10.17605/OSF.IO/GW5MX.

Michael Darius Eastwood. What Class of Corrector Is Deployed AI Alignment Actually Using? A Preregistered Classification Audit Against a Four-Cell Taxonomy. OSF registration, 2026j. Unit 06, the corrector-class audit; registered 1 October 2026, public from 4 October 2026. doi:10.17605/OSF.IO/RV2B5.

Michael Darius Eastwood. Unit 06 (corrector-class audit): results of the registered run, rv2b5. OSF, 2026k. Deposited 11 October 2026; the run of 6 October 2026, reported as administratively incomplete, with no inferential verdict. doi:10.17605/OSF.IO/4BPHG.

Michael Darius Eastwood. 82-study-cs--corrector-panel-independence · Testing the premise beneath Law III's value two (P17): do correctors drawn from one engine miss the same planted departures more often than independence allows? A counted test. OSF Registries, 2026l. Registered and public 27 October 2026. doi:10.17605/OSF.IO/JUFG4.

Michael Darius Eastwood. Testing Law V (the placement limb, P21): does correction inside each revision round pull away, as depth rises, from a monitor that sees only the finished register? A counted test. OSF registration, 2026m. Unit 83, the counted test of P21; submitted 22 October 2026, public from 25 October 2026. doi:10.17605/OSF.IO/MN73W.

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A Version history and corrections

This appendix keeps the paper's record whole. Section numbers inside the records are those of the version each describes; Appendix A.6 maps the sections of version 1.10 to this edition.

A.1 Version 1.12 (1 November 2026)

Version 1.12 gives this appendix's record of version 1.11 as one entry, where version 1.11 printed an earlier description beside later corrections, and gives each passage of version 1.10 that version 1.11 changed under its present place instead of an internal row label; it no longer prints wording from an unpublished draft of version 1.11. In the main text, wording about the paper's own preparation and approval gives way to plain statements (Sections 3 and 4.1, Remark 1, Section 6.4 and Appendix B), and the abstract closes on what the governor discussion does. Section 2.3 no longer quotes the author's December-era notes on Smolin, because the quotation's dated source has not been located; version 1.11 read:

> Smolin's cosmological natural selection (Smolin 1992) is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously ("Echoing Smolin's cosmological natural selection, AI could create recursive universes with their own laws", read at its source in the author's notes).

No result, equation, registration, source or date of a claim is added or moved, and the first-publication date remains 16 August 2026.

A.2 Version 1.11 (1 November 2026)

Version 1.11, 1 November 2026, set the paper on the programme's presentation standard. Its headline became a question, *When can correction keep pace with recursive self-acceleration?*, and its recorded title names what the threshold joins growth to, correctability, in place of "stability", the word retired from Paper X's title for the same criterion; until this version it read *Paper XIII · The ARC Theory · Laws I and II, the ARC Principle and the ARC Co-Scaling Law · The Self-Acceleration Exponent: a Measurable Threshold Joining Growth to Stability*. Every change below is carried in this one version.

- *Title block.* The first page carries the headline with its category and its place in the programme beneath it; the plate beneath the author line, now the first-page footnote, names Law II's correction criterion where it named Law II's stability.
- *Substantive: the correction criterion.* Section 4 defines alignment-stability as the relative misalignment D/C tending to zero, which in the standing gain-drift model holds exactly when $\beta_X > k$, and separates that relative criterion from absolute burden and harm; the abstract, Sections 6.4, 8, 8.1 and 10 and criterion F3 state the relative criterion where version 1.10 said stability. The gain-drift domain is specified, the strict super-exponential regime is distinguished from its exponential boundary, and Result 7's conjecture is carried into its heading, prose and table.
- *Substantive: estimates and measurement.* Paper II's error-decay estimates are separated from capability growth; its estimate of 0.31 is given as a parallel endpoint estimate, because the sequential and parallel arms were not matched in total compute; and the transformed error-suppression ratios are labelled hypothetical (Section 7.3). The negative-leverage extension is kept as mathematics, without an empirical substitution (Section 7.1). The self-acceleration exponent is the negative of the slope of cycle duration against capability, where version 1.10 called it the slope. Section 11 states what an admissible measurement needs and no longer calls it costless or its data already in hand, and no exponent is inferred from fixed rules or frozen weights (Sections 6.4 and 11).

- *Substantive: thresholds and overhead.* The ordering of the two thresholds, proportional verification overhead and additive latency are corrected in the current argument, including Result 8 and Figure 1; the nonshrinking-cycle cap is distinguished from the calibrated cap, which permits equality (Section 9.1).
- *Substantive: falsification.* The criteria separate mathematical counterexamples from empirical tests, state the limits of finite observation, and apply the current registration's equivalence, precision and replicated same-type P6 criteria (Section 12).
- *Substantive: status.* Section 14 gives every law's standing as at this version and Unit 83's result: P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (results: DOI 10.17605/OSF.IO/98HKR). It also gives Units 06 and 82 and the state of the measurement this paper defines, Unit 30; the status plate's note of 3 October 2026, the closing note after the references and the epistemic-status line are kept as printed in Appendix A.4.
- *Substantive: what other works said.* Section 2.2 gives Engels and co-authors their design condition in slope form and no longer calls the disagreement of the two programmes their most productive relationship; Section 2.3 states Hutter's undertaking without quotation marks, cites Chalmers, Good, Jones, Smolin and the threshold theorem, no longer says the literature contains no measurable exponent, describes the three impossibility papers as arguing for limits on alignment or its control, and gives the third's conclusion in its own words.
- *Substantive: the governor.* Section 10 is a scoped modelling contribution. Its comparison with primary sources distinguishes disturbance stability, variety, model-based regulation, goal-preserving self-modification, corrigibility and interruptibility from this paper's conditional exponent relation, and it claims no historical priority and no unique architecture; Section 2.1 states the contribution in the same terms.
- *Substantive: hypothesis and prediction.* Section 12.2 gives the human-in-the-loop prediction as a programme hypothesis beyond what this paper's algebra decides, as Paper X states it, and no longer says that it is stated here for the first time, since Paper X carried the same passage by 12 August 2026; Section 12.1 says that the same-class cap is the programme's conjecture, P6, which this paper does not derive.
- *Substantive: the pipes premise.* Section 3 states the fixed-substrate premise without claiming that software is the first or only system to improve at fixed throughput, gives the Eddington limit as a luminosity balance rather than a universal accretion ceiling, sets the idealised power laws beside the limits of a finite machine, and treats biological evolution as an analogy, not evidence.
- *Substantive: study AG.* Section 8.2 describes study AG as its registration text does: the two derivations share one functional form, and what it tests is their ordering.
- *Substantive: the substrate ceiling.* Section 8.1 states the additional premise behind the substrate-ceiling conjecture.
- *Substantive: control and usefulness.* Section 1 states the programme's position that people keep oversight, containment and shutdown for as long as they physically can, Section 10.2 points to it, and Section 9.1 gives the usefulness side of a cap on self-acceleration.
- *Editorial.* Sections 7.3 and 8.1 no longer use two phrases of the programme's working vocabulary, which Appendix A.5 gives as printed. The references are linked to their official sources and the programme's papers to their OSF DOIs; reference notes identify the editions consulted and distinguish their dates from first publication; the Ashby reference uses the trustee-authorised text, and the Wiener reference names the edition consulted and the publisher's reissue. The copyright block takes the reference edition's form, keeping the assertion of full human authorship; version

1.10's clause on the guidance of the United States Copyright Office and its link to the author's statement on authorship are not set, so that the only wording about tools is the author's disclosure; and the links about the author are his page and his book. Appendix A.5 gives every passage this version re-set as version 1.10 printed it, and Appendix A.6 maps the sections of version 1.10 to this edition.

No registered law, prediction, kill condition or dependency is changed by this paper, and no result number or empirical result is added. Current interpretations and Result 8's wording are corrected as stated above, without changing the dates of earlier claims. The 19 September 2026 notes and the other earlier words are kept below as history; they are not current claims. The first-publication date remains 16 August 2026.

A.3 Earlier versions, as printed

The version history of version 1.10, as printed, with each mention of a programme paper linked to its DOI.

Paper XIII v1.10, 5 October 2026: the title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Paper XIII · The ARC Theory · Laws I and II, the ARC Principle and the ARC Co-Scaling Law · The Self-Acceleration Exponent: a Measurable Threshold Joining Growth to Stability*; until this version it read *The Self-Acceleration Exponent: a Measurable Threshold Joining Law I's Growth to Law II's Stability (Paper XIII)*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. It says that the paper resolves the two notation collisions between the frameworks it joins, where it said the notation collision. The former subtitle beneath the plate, *A Measurable Threshold Joining Recursive Capability Growth to Corrective Stability*, which the title supersedes, is removed. Mentions of the programme's papers in this paper now carry their uniform titles. The note on how to read this paper now names the two laws the paper joins, Law I and Law II. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Paper XIII v1.9, 4 October 2026: the title now names the two laws the threshold joins, Law I and Law II; until this version it was titled *The Self-Acceleration Exponent*, with the subtitle *A Measurable Threshold Joining Recursive Capability Growth to Corrective Stability*. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Paper XIII v1.8, 4 October 2026: a dated note in the status box places this paper against Laws IV and V; it bears on Law V and not on Law IV, and no result in it is a deciding test of P15, P12 or P21. Substantive: the estimator row quoted from Paper II is now printed whole, once, in Section 5.1b, as one model's inconclusive fit, and that section says which estimate the interval belongs to; Sections 5.1, 5.1a, 5.2, 6.1 and 11 name the figure they use and point to that row, where they had printed the regression estimate beside the interval of the endpoint estimate or without its row; Section 14 names the pinned figures without that pairing, and the reference entry for Paper II names the row; the table in Section 5.1b keeps the interval's two ends, and Section 5.1a scopes sub-additive scaling to the point estimates of that fit. Substantive: Section 4.3a gives its reading of the adversarial-pressure result as a reading. Substantive: Sections 6.1 and 10.1 say preregistered by timestamp where they said registered, of the test of the identity and of the prediction on the ratio of the correction exponents, whose designs are drafts; a status note after Section 14 gives the state of the self-acceleration measurement on 3 October 2026. Substantive: study AE, since withdrawn from the programme, is no longer counted among the drafted registrations in Sections 2.1 and 6.1; words given in quotation marks in Section 13.2 that their source does not contain are replaced by what that source's abstract states; the closing note gives the state of the trials; and the epistemic-status line gives the standing of Laws I to III and of Laws IV and V separately. Substantive: Section 5.1b gives the count of models as Paper II gives it for its capability arm, four of five without a usable exponent where it had said five of six; the abstract names β'_L where it had named β_L among the quantities never measured; and Section 13 gives each work it credits in that work's own terms, cites the

1999 papers of West, Brown and Enquist and of Banavar, Maritan and Rinaldo, gives the 2010 paper of Banavar and co-authors the result it states rather than the general form, and names the basis of Zhao's derivation. Substantive: Section 7.2 and the reference entry for Paper VI give the honey-architecture runs as simulations and their constant advantage as a simulation result. A correction note in Section 5.1b records what was printed before, the table row taken out of that section included. References: the entry for Ball and co-authors gives the authors and the full title as arXiv records them, Section 13 names both authors of the 2010 paper by Demetrius and Tuszynski and cites the seven-author paper of 2010 as Banavar et al., the entry for Engels and co-authors prints the title as its registries give it, the Ashby entry says that no official online copy is available, the entries for Paper X and the statement paper carry the titles the programme's catalogue gives them, nine works that the text cites and the list lacked now have entries taken from their official records, and the works the paper cites are linked to their sources. Editorial: Sections 2.1, 13.2 and 14 name the author as the author; the heading of Section 2.1, the opening of Section 6 and four labels in Sections 13.1 and 13.2 are put in plain words; the paragraph of Section 6.1 headed In this paper's terms names study AG as the test of the identity without stating again the status the section gives above it; the version, the dateline and the metadata dates carried to v1.8. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Paper XIII v1.7: Four dated notes added after an owner review of the algebra in Sections 3, 4.4, 5.2 and 7.2, and no result is withdrawn. Substantive: a scope note beneath the fixed-tax paragraph in Section 7.2 records that the paragraph holds for a proportional verification overhead and not, in general, for an additive one, that an additive serial latency drives the local self-acceleration exponent to zero and removes the finite-time divergence wherever the untaxed model has one, for every β_L below 1, and that the honey-architecture retrodiction stands pending identification of what its tax models. Substantive: an ordering note beneath the two alarms in Section 5.2 records that their ordering is an observation about measured systems rather than a consequence of Result 3, because in the extended model of Section 5.1 the calendar-time threshold is $\beta_L + \delta = 1$ for every β_L below 1. Substantive: premise notes beside Result 1 in Section 3 and Result 2 in Section 4.4 state the finite-window solution and the aligned-origin or asymptotic condition under which the two accumulations may be treated as pure powers, and record that Result 3 is unaffected and is not withdrawn. Editorial: the version, the dateline and the metadata dates carried to v1.7. No reported number changes and the first-published date does not move.

v1.6 - 13 September 2026: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

v1.5 - 2 September 2026: seven duplicate machine citation blocks deleted. The block that remains now carries this paper's own DOI (10.17605/OSF.IO/HT8WU) in place of the programme umbrella, with a well-formed BibTeX key and a heading id used only once. Repository link repointed away from Paper XI's folder. Measured exponent now printed with its interval at every mention outside Section 5.1b. Engels et al. Spotlight status verified. Footer date of 25 August brought into line with the first-page date.

v1.4 - 27 August 2026: discoverability metadata and the 25 August figure and provenance notes.

A.4 Earlier notes, as printed

Four records of version 1.10 that this edition does not print in the main text, as printed: the status plate (its first paragraph is restated in Section 14.1), the correction note of 3 October 2026 from the section on the point estimate, the closing note after the references, and the epistemic-status line.

Status · a derivation, not a measurement This paper derives a measurable threshold, $\delta = 1/\alpha$, on a quantity any self-improvement loop can record, from one stated scope condition. It is not a measurement claim: three of the four quantities in the central result have never been measured in any system. Section

10 states what would refute it, section 12 records what earlier drafts withdrew, and section 13 states what is not claimed as new.

Two of the five conditions set for this paper's deposit remain open and are stated here rather than waived: the deciding measurement is drafted, dated and prepared as a draft registration awaiting human submission, and no mathematician without a stake in the programme has yet read the algebra adversarially. The author took the disclosure decision to publish on 16 August 2026 with both open conditions known. The programme-wide correction record is at /research/corrections/.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. Their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper bears on Law V. It argues, within its model, that correction embedded inside the recursive loop, accumulating as the loop runs, belongs to the only class of corrector whose correction exponent is a determined quantity (§4.4). Its elimination argument leaves one cell of its governor taxonomy: a governor that is part of the system and whose removal degrades the system (§8). It measures nothing, and that argument establishes only that the cell is the right one, not that any particular architecture successfully occupies it. It also gives, as a retrodiction, an account of a simulation result the programme had already published: in the complexity-scaling arm of the honey-architecture simulations the advantage of the entangled objective was constant rather than compounding (§7.2). That arm is Paper VI's, and it varied the network's complexity, not recursive depth. This paper reports no comparison of correction that takes part in each revision round with correction that sees only the finished output as depth rises, which is P21, and it says nothing about build order, which is P12. It does not bear on Law IV: its argument that a fixed rule catches less as capability rises (§4.3a) concerns a static corrector's correction exponent, not whether gains installed by external post-training persist under later capability-directed training, which is P15. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes no result printed here: the results, their dates and their limits stand as they were.

Correction note (3 October 2026). Until this version, sections 5.1, 5.1a, 5.1b, 5.2, 6.1 and 11 printed the regression estimate from Paper II beside the bootstrap interval that belongs to the endpoint estimate of the same row; section 5.1b said that the source reports the two together and set the regression estimate between that interval's two ends in its table, and section 14 described the pinned figures in the same pairing. Section 5.1b now prints the estimator row whole, as one model's inconclusive fit, its table keeps the interval's two ends, and the other passages point to the row. The row taken out of that table stood between the two ends: it gave the regression estimate, labelled as the point estimate, with 1.40 times per doubling of reasoning depth, a factor the paragraph above the table still gives. Section 5.1a gave sub-additive test-time scaling in frozen frontier models as the defensible claim; it now scopes that claim to the point estimates of that one fit. Section 4.3a said that a negative correction exponent is already measured in this programme, and that the measured static correctors fail before exponential growth; it now gives both as the framework's reading of the adversarial-pressure result it reports. Sections 2.1 and 6.1 counted study AE among the drafted registrations; it has since been withdrawn from the programme, and section 6.1 says so. Section 6.1 said that the identity is registered for testing, and section 10.1 that the prediction on the ratio of the correction exponents is registered; their designs, study AG and the corrector-class contrast (unit 01), are drafts preregistered by timestamp, first anchored on 12 August 2026 and on 28 August 2026, and both sections now say preregistered by timestamp. Section 13.2 gave, in quotation marks, words that the paper of Ball and co-authors does not contain; it now reports the

conclusion their abstract states. The closing note after the references said that nothing had been submitted; it now gives the state of the trials. The epistemic-status line said that what the programme names Laws are conjectures under registered adversarial test. It now gives the standing of Laws I to III and of Laws IV and V separately. Section 5.1b counted six frontier models tested, five of them yielding no usable exponent, and said that the benchmarks are too saturated on five of six. Paper II counts five models in its tier-2 capability arm, a sixth having been measured only in the alignment arm, and the section now gives four of the five. The abstract said that β_X and β_L have never been measured; it now names β_X and β'_L, the leverage fraction of correction, because §5.1a records that β_L has been measured in one regime. Section 13 listed five works together as independent derivations of the scaling exponent in its physical form. It now gives each work in that work's own terms: it cites the 1999 paper of West, Brown and Enquist beside their 1997 paper, and the 1999 paper of Banavar, Maritan and Rinaldo for the general relationship in networks; it gives the 2010 paper of Banavar and co-authors the result that paper states, an exponent of 2/3 when the velocity of flow is constant and of at most 3/4, where it had counted that paper among derivations of the general form; it gives the 2010 paper of Demetrius and Tuszynski in that paper's own terms, and Bettencourt's paper of 2013 the exponent that paper states for the land area of a city, $D/(D + H)$; and it names the basis of Zhao's derivation. Section 7.2 called the honey-architecture runs experiments and their constant advantage a negative result, and the reference entry for Paper VI called that advantage a result; the runs are simulations, and both now say so. Sections 2.1, 13.2 and 14 called the author the operator; they now say the author. The heading of section 2.1 said that ratification was pending and now says not yet adopted, the interpretation rule that opens section 6 is given with its date alone, and the four capitalised labels in sections 13.1 and 13.2 are now plain words. Unchanged: Results 1 to 8, the date of every claim, and the thresholds and falsification criteria as printed.

The measurement this paper defines is drafted, dated and prepared as a draft registration awaiting human submission, and it is preregistered by timestamp, first anchored on 12 August 2026 at block 962066; the full listing of the programme's drafted trials is public at the registered programme page. One study unit, the corrector-class audit (unit 06), was registered on OSF on 1 October 2026 and has been public since 4 October 2026 (DOI 10.17605/OSF.IO/RV2B5). Every other trial is a draft awaiting the author's submission, most of them preregistered by timestamp. The measurement this paper defines has not been run, and no result is claimed.

Epistemic status. What this programme names Laws are conjectures. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026; the predictions that carry them are registered, the laws themselves are not yet registered, and neither has been tested. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

Note (31 October 2026). The records above are kept as printed. The status plate's note of 3 October 2026 gives Laws IV and V the standing they had before 15 October 2026, when version 1.103 of the theory registration registered them as laws, and cites their mathematics at version 1.0; it now stands at version 1.2 of 29 October 2026. The epistemic-status line describes Laws I to III as under registered adversarial test and gives Laws IV and V their standing before 15 October 2026; each law is a law under test, and Law V's placement limb, P21, has a registered verdict from Unit 83. The closing note counts one study unit registered on OSF; three are registered, Units 06, 83 and 82. Section 14.3 gives each as at this version. Unchanged: the records above and every date in them.

A.5 Passages re-set in version 1.11

Each passage of version 1.10 that version 1.11 changed is given below as version 1.10 printed it, under its present place. The first list gives sentences and headings, each with the reason for its change; the second gives whole passages in the order version 1.10 printed them. Section numbers inside the quotations are those of version 1.10, and Appendix A.6 maps them to this edition. The quotations are the record of what was printed, including claims later withdrawn or narrowed; the current text governs.

Section 3 The example is not kept; the section states the fixed-substrate premise and its physical limits directly. Version 1.10 read: "Starve a tumour's vasculature and it stops."

Section 3 The Eddington limit is a luminosity balance, not a universal ceiling on accretion, and accretion above it is observed. Version 1.10 read: "Astrophysics has a named, quantitative version of the same ceiling: the Eddington limit, above which radiation pressure halts accretion."

Section 3 Evolution is treated as an analogy, not evidence, and no biological exponent is inferred; the theory registration places biological systems among those for which the framework offers an analogy and no measurement. Version 1.10 read: "On this framework, evolution should carry a measurable exponent far below the ceiling; that is a prediction about biology falling out of a framework built for software, and it is stated as one."

Section 3 The section does not claim that software is the first or only system to improve at fixed throughput, and it sets the idealised power laws beside the limits of a finite machine. Version 1.10 read: "Software is the first thing that keeps growing when you hold the pipes fixed."

Section 5 The section no longer says that the clock relation is required by Paper X's speed-prefactor result, which varies b at fixed exponents in its stated drift domain; neither fact makes α a mere speed coefficient. Version 1.10 read: "That is not merely consistent with the corrective framework's speed-independence theorem; it is required by it,"

Section 6.4 A rule being fixed no longer establishes $\beta_X \leq 0$; the stricter relative criterion is stated only for a mechanism whose strength decreases. Version 1.10 read: "So class 1 gives $\beta_X \leq 0$, and stability requires"

Section 7.3 The sentence is not kept; the section gives the estimates with their precision and labels the transformed ratio hypothetical. Version 1.10 read: "it is the most communicable figure in the estate,"

Section 7.3 The sequential and parallel arms were not matched in total compute, so 0.31 is a parallel endpoint estimate. Version 1.10 read: "a compute-matched parallel exponent of 0.31"

Remark 4 For $\beta_X > k$ the relative misalignment vanishes on either side of the growth boundary, while absolute misalignment may still grow. Version 1.10 read: "but a system past it with $\beta_X > k$ remains stable."

Remark 4 Neither boundary is a general safety threshold; the upper one decides only whether the relative misalignment vanishes. Version 1.10 read: "the stability question is decided at the upper boundary alone."

Section 8 The section no longer contrasts fast and slow takeoffs: takeoff speed alone does not determine the relative outcome, and a low relative error does not establish safety. Version 1.10 read: "can cross the stability boundary at modest δ."

Section 8.1 A word of the programme's working vocabulary is replaced. Version 1.10 read: "The substrate-ceiling conjecture and the width identity combine, and neither lane derived this alone."

Section 8.1 Within S1 and the ceiling conjecture, $k < 1$ is necessary, not sufficient, for vanishing relative misalignment. Version 1.10 read: "And stability then requires $k < 1$"

Section 8.2 The words described the withdrawn draft of study AG; its registration text says that the two derivations share one functional form, and that what it tests is their ordering. Version 1.10 read:

> Study AG tests the two derivations of the critical exponent against each other on the same system, notes that they agree numerically only at the single point where both parameters equal one half and disagree in their derivatives everywhere else, and is designed to find the disagreement, not to confirm the agreement.

Section 10 The comparison $\beta_X - k$ decides whether the relative misalignment vanishes, within the stated drift model; it is not a characteristic root, and Maxwell's stability keeps its classical sense. Version 1.10 read: "sign$(\beta_X - k)$ is a stability verdict from an exponent comparison,"

Heading of Section 11 The heading no longer calls the measurement costless. Version 1.10 read: "9. The measurement that costs nothing"

Section 11 The section no longer says that the exponent rises first or that its data are already in hand: no algebra forces it to rise before a correction failure, and whether existing records suffice is a condition to check. Version 1.10 read: "this is a quantity that rises first, from data already in hand."

Criterion F3, Section 12 The criterion separates a mathematical counterexample from an empirical test; a fraction that falls over a finite window does not refute the asymptotic boundary. Version 1.10 read: "that remains stably correctable over a sustained observation window,"

Criterion F3, Section 12 A counterexample satisfying the standing assumptions would refute the claimed asymptotic boundary for vanishing relative misalignment. Version 1.10 read: "The stability boundary is wrong."

Section 12.1 The cap is the programme's conjecture, P6 of the theory registration, which this paper's algebra does not derive. Version 1.10 read: "This framework predicts it is capped."

Heading of Section 12.2 The prediction is a programme hypothesis beyond what this paper's algebra decides. Version 1.10 read: "10.2 A second prediction, free, from the same mechanism"

Section 12.2 It is a programme hypothesis beyond what this paper's algebra decides, and it was not stated here for the first time: Paper X carried the same passage by 12 August 2026, before this paper's first publication on 16 August 2026. Version 1.10 read:

> The same mechanism yields a second measurable prediction, stated here for the first time: oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The framework therefore predicts that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.

Section 2.2 Engels and co-authors are given their design condition in slope form, and the disagreement of the two programmes is stated at its size. Version 1.10 read:

> The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. Their framework contains no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. Nested Scalable Oversight is iterated same-class oversight by construction, and this programme's central prediction is that the same-class ladder is bounded however many rungs are added, while a cross-class corrector is not. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have.

Section 2.3 Hutter's undertaking is stated in his own terms, the works the paragraph names are cited, and the paragraph no longer says that the literature contains no measurable exponent. Version 1.10 read:

> On the question. Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", arXiv, 28 February 2012, read at its source), separating "speed from intelligence explosion" and undertaking to "consider possible bounds on intelligence", augmenting Chalmers' 2010 analysis. The question and the speed-versus-structure distinction are therefore at least fourteen years old. What that literature does not contain is a number: no measurable exponent, no derived ceiling, no architecture dependence.

Section 2.3 The three works are described as arguing for limits on alignment or its control, and the third's conclusion is given in its own words. Version 1.10 read:

> On impossibility. Three 2025 arXiv papers argue that perfect control is unattainable: Yao, "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, v1 12 June 2025 02:30:30 UTC, a time taken from arXiv's record alone, independently observed by the Internet Archive on 13 June 2025; cited by arXiv:2512.03048); Yao, "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, the only paper in the arXiv abstract corpus containing the phrase "irreducible uncontrollability", abstract-search total of one, measured 12 August 2026); and Ball, Gluch, Goldwasser, Kreuter, Reingold and Rothblum, "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The third, notably, concludes that safety cannot be achieved by designing filters external to a language model's internals (architecture and weights), an independent argument, from filtering intractability, in the same direction as this programme's architecture dependence; it is convergent support on that leg, not a rival. Two of the three papers are by one sole author; the description "a wave" overstates the literature's breadth, though not the seriousness of the six-author paper.

Section 2.3 The threshold theorem is cited. Version 1.10 read: "The quantum error-correction threshold theorem also converts"

Section 2.3 Smolin's work is cited. Version 1.10 read: "Smolin's cosmological natural selection is the antecedent"

Title block The presentation standard puts the category and the paper's place in the programme beneath the title. Version 1.10 read: "The ARC Theory (the Theory of Artificial Recursive Creation) · ARC/Eden experiments · Paper XIII · Working Paper v1.10"

Section 14.3 The registration now stands at version 1.104. The plate's links to the registration's formats and its OSF record, labels of the website, are not repeated here. Version 1.10 read:

> Theory-level predictions preregistered on OSF: DOI 10.17605/OSF.IO/P8CKQ, registered 8 September 2026 (version 1.100), updated 13 September 2026 (version 1.102), twenty-two propositions.
>
> Companion: Register · The ARC Theory · Operational Definitions, version 1.9.1, 5 October 2026.

Author block and footnote The presentation standard sets the author block, and the links about the author are his page and his book. Its last item, a label linking to a page of the website, is not repeated here. Version 1.10 read:

> Michael Darius Eastwood
> Independent AI alignment researcher, London · Author, *Infinite Architects* (2026)
> ORCID 0009-0004-3222-7442 · michael@michaeldariuseastwood.com · michaeldariuseastwood.com
> The ARC Theory · OSF osf.io/ht8wu · [. . .]

First-page footnote The footnote names Law II's correction criterion where the plate named Law II's stability. Version 1.10 read: "The bridge between Law I's growth and Law II's stability; it also resolves the two notation collisions between the frameworks it joins, and it bears on Law V."

Footnote and version note The presentation standard sets the version, the dates and the DOI in the footnote. Version 1.10 read: "Working Paper v1.10 · First published 16 August 2026, revised 5 October 2026 · DOI: 10.17605/OSF.IO/HT8WU"

Section 14.1 The status of each result is now stated once, here, together with the passage given next. Version 1.10 read:

> How to read this paper. This paper joins the scaling framework of Law I, the ARC Principle, to the corrective framework of Law II, the ARC Co-Scaling Law. Claims come in three sizes: results, laws, frames. This is a laws-and-frames paper: Results 1 and 3 are algebra, Results 2, 5 and 6 are algebra given a stated scope condition, and Result 7 additionally rests on a conjecture that is flagged rather than folded in. Nothing here is measured, and the paper says so wherever the temptation to imply otherwise would arise.

Abstract The presentation standard sets the why first, in at most 1,920 characters, and Law II's criterion is stated as the relative misalignment tending to zero. Version 1.10 read:

> Two frameworks in this programme describe recursive self-improvement and have never been related. One derives how capability compounds with recursive depth and yields the scaling exponent $\alpha = 1/(1 - \beta_L)$. The other derives when a self-improving system remains correctable and yields the criterion $\beta_X > k$. They use the same two letters for different quantities, which has prevented the comparison rather than merely obscuring it.
>
> This paper resolves the notation, derives the relation between the frameworks, and finds that the quantity deciding the outcome is neither capability nor speed but how fast a system compresses its own iteration time. Writing cycle duration as $\Delta t \propto C^{-\delta}$ defines the self-acceleration exponent δ, and the growth exponent of the corrective framework follows exactly:
>
> $$k = \delta - 1/\alpha$$
>
> The threshold is $\delta = 1/\alpha$. Below it a system's capability grows as a power law and the corrective question does not arise. Above it growth is super-exponential and correction must out-scale it. On one assumption about the form of the correction process, the survivable window is
>
> $$1/\alpha < \delta < 1/\alpha + \beta_X$$
>
> two boundaries whose gap is exactly the correction exponent. Only the upper one is a stability threshold: the lower marks where growth leaves the power law, and a system between them is in a hard takeoff and still correctable, which contradicts the standard picture in both directions.
>
> δ requires no new instrument, benchmark or model access: it is the slope of cycle duration against capability on logarithmic axes, on a fixed substrate, from series any self-improvement loop can record out of what it already computes, though, as §9 records, the two series are not currently paired anywhere. β_X and β'_L have never been measured; δ has not either, and it is the one that could be measured within a month of a loop existing.

Section 14.1 Stated once, with the passage above; Section 2.1 points to it. Version 1.10 read: "Claimed with the status it has. Results 1 and 3 are algebra. Results 2, 5 and 6 are algebra given S1's form condition. Nothing here is measured."

Whole passages. The passages below are quoted from version 1.10 (5 October 2026). Their mathematical symbols are set from a plain-text reading of that version.

Section 1 "Whether a self-improving system remains correctable has been argued for two decades in words. The positions divide on takeoff speed, and both sides treat the outcome as decided by a level : how capable the system becomes, or how fast it gets there."

Section 1 "This paper argues the outcome is decided by neither, and identifies the quantity that decides it. The argument requires joining two frameworks that were built independently and cannot currently be compared, because they collide in notation."

Section 3 "Everything that grows is fed through a channel, and for everything before software, fixing the channel fixes the growth. Starve a tumour's vasculature and it stops. Fix a chain reaction's fuel and geometry and it stops. Exhaust an epidemic's susceptible hosts and it burns out. Astrophysics has a named, quantitative version of the same ceiling: the Eddington limit, above which radiation pressure halts accretion. The claim here is a generalisation of limits physics already accepts, not a new species of assertion."

Section 3 "Two hard cases are handled deliberately. Cosmic inflation is excluded by scope: it is expansion of space, not growth of a structure on a substrate, and it ended by field dynamics rather than by running out of anything. Evolution is the sharpest case and it fits: biological complexity increased for billions of years at roughly fixed solar throughput, which is growth on information at fixed pipes, and it is also glacial, and it has no internal corrector at all, since selection is external. On this framework, evolution should carry a measurable exponent far below the ceiling; that is a prediction about biology falling out of a framework built for software, and it is stated as one."

Section 3 "Software is the first thing that keeps growing when you hold the pipes fixed. The operational definition does the work here and is load-bearing, not a technicality: an artefact loop on a fixed substrate, where each pass works on the accumulated output of the previous passes, measured on a substrate-fixed clock. A model merely thinking longer does not qualify. Growth that only appears when the substrate is enlarged is pipe-fed growth and stays under the old law; growth at fixed substrate is the new regime, and it is the regime this programme's limit addresses."

Section 4 "The consequence of crossing the β collision. Substituting $\beta_L = 0.5$ from the scaling framework into the corrective criterion sign(β_X − k) is not an approximation. By §4 it corresponds to β'_L = 0, a corrector with no leverage at all: one that cannot use its own accumulated corrections. The substitution silently assumes the least capable corrector constructible and presents the result as a property of the system."

Section 5 "Consequence. The corrective framework's control quantity q = A/r_rate ∝ C^{β_X − k} becomes, in recursion depth, q ∝ r^{$\alpha(\beta_X$ − k)}. So α amplifies the rate at which the verdict arrives per recursive step, and cannot change the verdict. That is not merely consistent with the corrective framework's speed-independence theorem; it is required by it, since α is a speed in the only sense that matters, and a change of variables cannot flip a sign."

Section 6.2 "The taxonomy closes as a decision tree over what A is permitted to depend on , which is exhaustive because those are the only available arguments: nothing, capability, or its own history, and the last splits by substrate."

Section 6.2 "A depends on class β_X nothing 1 · static. Fixed rule, filter, written constitution, guardrail 0 at best; ≤ 0 in practice (§4.3a) capability only 2 · capability-carried. A larger reviewing model does the checking free primitive. Chosen by provisioning, at the cost of a regress its own history, same substrate 3 · self-accumulating determined by Result 2 below. The only class in which β'_L has a referent its own history, other substrate 4 · cross-class may exceed 1. The one route out of the ceiling"

Section 6.2 "Correspondence with the governor taxonomy of §8 is exact : class 1 is Maxwell's fixed flyballs, class 2 is Ashby's external regulator, class 3 is the cell empty since 1868 , and class 4 is the composition-class escape. This is what closes the exhaustiveness objection §8's elimination argument would otherwise face , and the two-binary version did not."

Section 6.2 "It also determines the design of the class audit. The audit's whole task is to separate class 1 from class 2, and under the two-binary scheme it could not have."

Heading of Section 6.3 "4.3 Result 2a · a static corrector fails at exponential growth, and this needs no conjecture"

Section 6.3 "For a class 1 corrector, $\beta_X = 0$ at best , because a strength depending on nothing is A_0C^0. Reading an exponent off a constant, not an estimate."

Section 6.3 "and $\dot{C}$ = bC^{1+k} with k < 0 is sub-exponential. So, stated with nothing beyond the corrective framework itself:"

Section 6.3 "A static corrector is adequate only while capability growth is sub-exponential. At exponential growth and above, it fails."

Heading of Section 6.4 "4.3a And $\beta_X = 0$ is the generous case, not the value"

Section 6.4 "A fixed rule against an improving system does not hold constant. It catches less as capability rises, because the system routes around it. That is $\beta_X < 0$."

Section 6.4 "And, on the framework's reading, it has already been observed in this programme. The adversarial-pressure test found ethics collapsing while capability held or improved, with Pearson r between −0.980 and −0.999 in three of six frontier models. A corrective mechanism whose effect degrades as capability rises has a negative correction exponent by definition."

Section 6.4 "So class 1 gives $\beta_X \leq 0$, and stability requires"

Section 6.4 "which is stricter than k < 0."

Section 6.4 "Stating Result 2a as a best case is what makes it strong. Even granting static correctors the most generous assumption available to them, they fail at exponential growth. On the framework's reading, the measured ones fail before that. Every filter, fixed rule, written constitution and static guardrail is class 1, and the class of intervention on which most current practice rests has a hard expiry condition rather than a weakness."

Section 6.4 "Class 2 does not escape by having a free β_X. Its freedom is purchased by provisioning a checker that is itself capable, which relocates the problem rather than solving it. That is the regress the corrective framework 's antecedent-work section credits as scalable oversight, and §8's elimination argument applies to it as an external governor."

Section 6.5 "The consequence for practice is the paper's central engineering claim. Correction embedded inside the recursive loop, accumulating as the loop runs, is a class 3 corrector. Class 3 is the only class in which the correction exponent is a determined quantity rather than a hope. An entangled objective, in which the safety term is part of what the loop optimises and compounds with it, is class 3 by construction."

Section 7 "The substrate qualifier is load-bearing, not decorative. Wall-clock cycle time also falls when operators buy faster hardware or optimise the inference stack, and that shortening is a change in the coefficient b, which the verdict does not contain. Only compression the system produces for itself belongs in δ. A measurement that pools the two will report $\delta > 0$ for nearly every deployed system and mean nothing by it, and it would contaminate F4 below: exogenous speed-up can mimic super-exponential wall-clock growth at fixed-substrate $\delta = 0$, which would read as a refutation of Result 3 and be a procurement decision. Every quantitative claim in this paper is in the substrate-fixed clock."

Section 7 "δ k regime 0 $-1/\alpha < 0$ constant cost per step. Capability grows as a power law $< 1/\alpha < 0$ still power law. The corrective question does not arise = $1/\alpha$ 0 exponential in time. The threshold $> 1/\alpha > 0$ super-exponential. Correction must out-scale it"

Heading of Section 7.1 "5.1 Negative leverage, and why no measured system is near the threshold"

Section 7.1 "The framework restricts β_L to [0,1), and the identity therefore yields $\alpha \geq 1$. Yet the best cross-architecture measurement the programme has taken, a regression estimate of $\alpha \approx 0.49$ from one model's inconclusive fit (its row is printed whole in §5.1b), falls outside anything that range permits. This is an internal inconsistency and it must be resolved before any threshold is quoted

numerically."

Section 7.1 "It resolves in the identity's favour, not the range's. The identity extends to $\beta_L < 0$ without modification:"

Section 7.1 "dg/dr = a·g^(β_L), $\beta_L < 0 \implies$ g ∝ r^(1/(1−β_L)), $\alpha < 1$"

Section 7.1 "The ODE remains well posed. $\beta_L \approx -1.04$ gives $\alpha = 1/2.04 \approx 0.49$, matching the regression estimate."

Section 7.1 "And negative leverage has a clear physical reading: each step uses less of the accumulated context than the step before it. Sub-additive returns to accumulated reasoning, which is what a frozen model with a fixed context window and no weight update would be expected to show."

Section 7.1 "So the measured value is not inconsistent with the framework. It is inconsistent with the framework's assumed range , and the range should be widened to $\beta_L \in (-\infty, 1)$ with three named regimes:"

Section 7.1 "β_L α reading < 0 < 1 sub-additive. Each step leverages less than the last = 0 1 additive. Steps are independent (0, 1) > 1 super-additive. Genuine compounding"

Heading of Section 7.2 "5.1a What the measurement is a measurement of, scoped precisely"

Section 7.2 "Of the frozen frontier models tested, their weights held fixed, one gave the regression estimate $\alpha \approx 0.49$ when capability was regressed against reasoning depth; that model's fit is inconclusive, and its row is printed whole in §5.1b. It is a measurement of test-time compute scaling. It is not a measurement of recursive self-improvement, and it must never be presented as one."

Section 7.2 "A frozen model reasoning for longer is not improving itself. Its weights do not change, its attention patterns do not reorganise, and its reasoning rules do not rewrite themselves. Nobody in the field has claimed that chain-of-thought is recursive self-improvement , so a sub-additive result on chain-of-thought cannot contradict any claim about RSI. An earlier draft of this section implied that it did. That inference is withdrawn."

Section 7.2 "The defensible claim, and it is worth having on its own terms:"

Section 7.2 "On the point estimates of the one frontier model that gave a usable fit, test-time compute scaling is sub-additive; the fit is inconclusive. On those estimates the leverage exponent is negative: each additional increment of reasoning depth uses less of the accumulated context than the increment before it."

Section 7.2 "That bears directly on the "let it think for longer" thesis , which is a live and well-resourced position, and it is a claim about a regime that has actually been measured. Dressed as a claim about recursive self-improvement it is refutable in one sentence by anyone at a laboratory, and the refutation would be correct."

Section 7.2 "So the honest status of β_L is that it has been measured in one regime and not in the regime the framework is about. The first threshold, β_L crossing zero, remains unmeasured for any system that modifies its own weights while improving, which is why that measurement is the highest-value one available and is not the one this paper's instrument addresses."

Heading of Section 7.3 "5.1b The point estimate must not be quoted, and the interval is the finding"

Section 7.3 "The regression estimate, $\alpha \approx 0.49$, offers up a tempting sentence: double the reasoning depth and capability rises by 2^0.49 ≈ 1.40 times, well short of double. It is arithmetically correct, it is the most communicable figure in the estate, and it must not be published."

Section 7.3 "On Gemini 3 Flash, the source, Paper II , reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of [−1.3, 2.9], a regression sequential exponent of 0.49 with a

standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31. It is one model's fit, and it is inconclusive. The interval belongs to the endpoint estimate, not to the regression estimate. Carried through the same exponentiation, its two ends give:"

Section 7.3 "α doubling reasoning depth buys −1.3, interval floor 0.41× , thinking longer makes it worse 2.9, interval ceiling 7.46×"

Section 7.3 "One model's inconclusive fit, whose endpoint interval spans degradation to seven-and-a-half-fold, cannot support a headline of 1.40. This is the retracted $\alpha \approx 2.24$ error in the same shape, and the programme has already written the rule that forbids it: the earlier interval "was wide enough to be consistent with both the theory and its negation, and should not be treated as confirmatory of either." That rule binds here."

Section 7.3 "The publishable claim is about the precision of the evidence, and it is stronger than the point estimate would have been."

Section 7.3 "Of the five frontier models that carried the tier-2 capability arm of Paper II (a sixth entered the study and was measured only in the alignment arm), four yielded no usable exponent at all : one reached ceiling at 94 to 100 per cent, one produced a binary step function, one scored 100 per cent at every depth, one showed no trend. Only one model produced clean monotonic data , and the bootstrap interval on its endpoint estimate is [−1.3, 2.9]."

Section 7.3 "The scaling of capability with reasoning depth has not been measured to useful precision on any frontier model. On four of the five the benchmarks are too saturated to measure it at all, and on the fifth the interval cannot distinguish degradation from a sevenfold gain per doubling."

Section 7.3 "That claim does not depend on the point estimate , so it cannot be refuted by a laboratory reporting a different one. It invites them to produce a tighter interval, which is the response worth provoking. And it is a second instance of the saturation thesis this programme already argues elsewhere : benchmarks saturated past the point of measuring what is being claimed about them."

Appendix A.4 Version 1.10's correction note of 3 October 2026, from its Section 5.1b, is printed whole in Appendix A.4.

Heading of Section 7.4 "5.2 Two thresholds in sequence, and the first one is not δ"

Section 7.4 "The thresholds are α-dependent and quoting 0.5 without its α is an error. Take the regression estimate $\alpha \approx 0.49$, from one model's inconclusive fit (§5.1b), and the self-acceleration threshold comes out at $1/\alpha \approx 2.04$, nowhere near 0.5. A sub-additive system needs enormous cycle compression before its growth leaves the power law , which is a reassuring result and follows directly."

Section 7.4 "This puts a second, earlier threshold ahead of δ:"

Section 7.4 "β_L crosses 0. Leverage becomes positive, α exceeds 1, and the system begins genuinely compounding. No measured system has crossed this. It is the first alarm, and it is estimable from the same recursive-depth experiments that already estimate α. δ crosses $1/\alpha$. Self-acceleration outpaces the compounding and growth becomes super-exponential. Only meaningful once (1) has happened , and the threshold moves as α moves."

Section 7.4 "Ordering note (19 September 2026). An owner review of the two thresholds above finds that their ordering is an observation about measured systems and not a consequence of the algebra, and this note states which it is. In the extended model of §5.1, where β_L is permitted to be negative, the calendar-time threshold is $k = 0$, that is $\beta_L + \delta = 1$, for every $\beta_L < 1$, so positive leverage is not a precondition for crossing it. Taking the equation values $\beta_L = -1$, whence $\alpha = 1/(1 - (-1)) = 1/2$, together with $\delta = 5/2$, gives $k = 5/2 - 2 = 1/2 > 0$, and by Result 3 the calendar-time growth is

proportional to C^(1 + k) = C^(3/2): per-step leverage is strongly negative and per-step growth is sublinear, yet the calendar-time threshold has been crossed. Those numbers are equation values chosen to display the point. They are not a measurement, a parameter estimate or a claim about any system. What this note restricts is the word "only" in the second alarm, and not the alarms themselves. Unchanged: k = $\delta - 1/\alpha$; the first alarm's estimability from the same recursive-depth experiments; and the statement that no measured system has crossed $\beta_L = 0$. Read as a statement about the systems this programme has measured, the sequence above stands. Read as an implication of Result 3, it does not."

Section 7.4 "So the correct reading of the framework's present status is not that the danger is near. It is that the first of two thresholds has not been crossed by anything yet measured, and that both are measurable. The $\delta = 0.5$ figure applies to a system at $\alpha = 2$, which is a candidate ceiling and not an observed value for any system."

Section 8 "Interpretation rule, added 8 August 2026: this section names two boundaries and only one of them is a safety threshold. The verdict is sign(β_X − k) and nothing else flips it. Crossing k = 0 (the lower boundary, $\delta = 1/\alpha$) changes the growth regime and collapses the wall-clock time an operator has to react, which the corrective framework itself notes is operationally real, but a system past it with $\beta_X >$ k remains stable. Any reading of this section as "inside the window is dangerous" or "the window is where safety lives" is wrong in both directions: the stability question is decided at the upper boundary alone. The earlier drafting called this a corridor, which implied both edges mattered equally, and that framing is retired."

Section 8 "The lower edge is where growth leaves the power law. The upper edge is where correction can no longer out-scale it. Between them a system is in a hard takeoff and still correctable."

Section 8 "The width of the survivable hard-takeoff window is exactly the correction exponent. In the neutral case $\beta'_L = \beta_L$ it is exactly 1, independent of β_L, so the boundaries' position moves with the scaling exponent while the gap between them does not."

Section 8 "This is the paper's engineering result. Improving correction does not raise a ceiling on capability and does not slow the system. It widens the window of self-acceleration the system can survive, one-for-one. A system with $\beta_X = 1$ tolerates a unit of self-acceleration above threshold; a system with $\beta_X = 2$ tolerates two."

Section 8 "It contradicts the standard picture in both directions. A fast takeoff is not automatically a lost one: a system well past the lower boundary is accelerating hard and remains correctable. A slow takeoff is not automatically safe: a system with weak correction has a narrow gap and can cross the stability boundary at modest δ."

Heading of Section 8.1 "Result 7 · the gap can never be wider than one"

Section 8.1 "with equality exactly when $\beta'_L = \beta_L$. This is not a bound on a number so much as a statement about what is constructible:"

Section 8.1 "The best possible internal corrector scales linearly with capability. It can never out-scale it."

Section 8.1 "However well the corrector is built, the survivable window of self-acceleration is at most one unit of δ wide, and it sits between $1 - \beta_L$ and $2 - \beta_L$."

Section 8.1 "β_L power law ends, k crosses 0 beyond any internal corrector 0.25 $\delta >$ 0.75 $\delta \geq$ 1.75 0.50 $\delta >$ 0.50 $\delta \geq$ 1.50 0.75 $\delta >$ 0.25 $\delta \geq$ 1.25"

Section 8.1 "The two thresholds are exactly one apart at every value of β_L. The position of the window moves with the scaling exponent; its width does not."

Section 8.1 "And stability then requires k < 1 : a universal threshold at k = 1, which is $\dot{C} \propto C^2$, independent of where the ARC Bound sits."

Section 8.1 "Status. Result 7 depends on the substrate-ceiling conjecture, which is prepared as a separate draft registration awaiting human submission and is not assumed elsewhere in this paper. Results 5 and 6 stand without it. Two unproven conjectures multiplied together is the failure mode this programme has already suffered once, and Result 7 is flagged rather than folded into the main line for exactly that reason."

Section 8.2 "Surfaces therefore say: the identity is a claim preregistered by timestamp for testing, the deciding measurements are drafted and dated, and if the channels disagree the unification is withdrawn from every surface it has reached, with the same prominence as the framing. Surfaces never say the identity is established, and never present the agreement of an asserted one half with the regression estimate of 0.49 (§5.1b) as evidence, because the two may share an accumulation law and agreement between sharers is not corroboration."

Section 8.2 "In this paper's terms. The clock relation k = $\delta - 1/\alpha$ (Result 3), the ratio $\beta_X = (1 - \beta_L)/(1 - \beta'_L)$ (Result 2) and the width identity boundary gap = β_X (Result 6) are algebra given the definitions of §4 and §5, and their status is stated as algebra in §13. The identity between the programme's separately derived critical exponent and this paper's threshold $1/\alpha$ is governed by the paragraph above: study AG is its test, it is not asserted as established here, and the deciding measurements are prepared as draft registrations awaiting human submission. For the same reason, any agreement between the asserted 1/2 and the regression estimate $\alpha \approx 0.49$, whose row is printed whole in §5.1b, is not put forward here as corroboration."

Section 9 "The criterion contains no capability level and no speed. α, β_X, β'_L and δ are all exponents; the coefficients that carry units of time appear nowhere in the verdict. This is the corrective framework's speed-independence theorem restated in the joined variables, and it has a consequence worth naming plainly."

Section 9 "If the criterion holds, there is no capability level at which correctability becomes impossible. A ratio between two exponents can be maintained at any absolute scale. The question is never how capable a system has become; it is whether its capacity to correct itself has kept pace with its capacity to improve itself."

Section 9 "Both the position that recursive self-improvement is inevitably uncontrollable and the position that it is inherently safe assume the outcome is decided by a level. On this account neither follows."

Heading of Section 9.1 "7.1 Result 8 · one rate-like intervention does work, and it is not the one being used"

Section 9.1 "By Result 3, k = $\delta - 1/\alpha$. δ is inside k. So the verdict sign(β_X − k) is not indifferent to every rate-like quantity. It is indifferent to exactly one of them."

Section 9.1 "intervention acts on effect on the verdict cap throughput: fewer FLOPs, smaller runs, a moratorium the coefficient b in $\dot{C}$ = bC^{1+k} none. b does not appear in the verdict cap self-acceleration: bound how fast a system may shorten its own cycle δ, inside k direct. Reducing δ reduces k one-for-one"

Section 9.1 "Throughput caps cannot work. A cap on self-acceleration can, and it is the only rate-like intervention the algebra permits."

Section 9.1 "This is actionable without any measurement, and the strongest form of it needs no parameter at all. From k = $\delta - 1/\alpha$: enforce $\delta \leq 0$, that is, a self-improvement loop's cycle time may never shrink, and k $\leq -1/\alpha < 0$ for every $\alpha > 0$. Within the model, "the loop may never speed up" guarantees the sub-exponential regime unconditionally, with nothing estimated, for any architecture, at any capability. δ is observable on a fixed substrate, attributable to the operator who runs the loop, and enforceable by construction."

Section 9.1 "A calibrated version exists for operators willing to measure: a minimum cycle duration shrinking no faster than C^{-1/α} permits self-acceleration up to the threshold. But the calibrated cap is indexed to α, which no regulator can observe and no measurement has yet pinned, so the unconditional form is the governable one."

Heading of Section 9.2 "7.2 And it explains a negative simulation result this programme already published"

Section 9.2 "Stated as a retrodiction, not a prediction. The result was known before the derivation existed."

Section 9.2 "A fixed tax adds a constant to cycle duration. A constant changes b. b does not appear in the verdict. So a fixed verification tax cannot move the outcome, and an advantage that does not compound is what that looks like from outside."

Section 9.2 "Scope note (19 September 2026). An owner review of this section's algebra finds that the paragraph above holds for a proportional verification overhead and not, in general, for an additive one, and §10 and §11 of this paper already imply the distinction. Write the taxed step duration as $\Delta t_h = b\, C^{(-\delta)} + h$, with $h > 0$ a serial latency that cannot be parallelised away. The local self-acceleration exponent, defined as $-d \log \Delta t_h / d \log C$, is then $\delta\, b\, C^{(-\delta)} / (b\, C^{(-\delta)} + h)$, which decays to zero as C grows instead of holding at δ. With $dr = dC / (a\, C^{(\beta_L)})$, the elapsed time to capability C acquires the further term $(h/a) \int u^{(-\beta_L)}\, du$, whose antiderivative $u^{(1-\beta_L)}/(1-\beta_L)$ diverges at the upper limit for every $\beta_L < 1$; so within the bound $\beta_L \in [0,1)$ declared in §2, a positive additive latency removes the finite-time divergence in every region where the untaxed model has one, including the whole of the super-exponential regime $k > 0$. A proportional overhead $\Delta t_q = q \cdot b\, C^{(-\delta)}$ leaves δ untouched and changes only b, so for a proportional tax the paragraph above is exactly right; so is the additive case at $\delta = 0$, where b simply becomes b + h; and over a bounded range with h much smaller than Δt a coefficient-only reading may still be defensible if the range and the error are quantified. What this note restricts is the word "fixed" taken without qualification. F4 in §10 and the paragraph in §11 stating that iteration time may not follow a constant-exponent law both record that Result 3 is algebra given a single constant exponent, and an additive latency is precisely a departure from that parameterisation. Unchanged: Result 3 itself, the clock relation of §3, and the §7.1 argument that enforcing $\delta \leq 0$ buys the sub-exponential regime unconditionally, which the additive result strengthens, since an enforced positive serial latency is a second and constructive route to the same guarantee. The honey-architecture paragraph above is a separate matter: this paper does not establish that the honey-architecture tax is an additive per-cycle wall-clock constant rather than a penalty applied to a combined score, and the retrodiction stands pending that identification."

Section 9.2 "The same algebra specifies the arm that would differ: a verification cost that scales with capability , which acts on δ rather than on b. That arm has never been run, in this programme or elsewhere, and it is prepared as a separate draft registration awaiting human submission."

Section 9.2 "Three lines of work meet at δ, and none of them was aimed there. The engineering programme arrived at capability-scaled verification cost; the policy analysis arrived at the only governable rate quantity; the algebra arrived at the term inside k. They are the same quantity."

Heading of Section 10 "8. The governor, and the cell that has been empty since 1868"

Section 10 "The mechanism this criterion describes is a governor : not a limit the system must stay under, but a restraining mechanism whose force rises with the output it restrains. In Watt's centrifugal governor the flyballs spin faster as the engine does and close the throttle by the same motion, so the restraint is not applied to the engine, it is part of the engine and scales with it."

Section 10 "This is not a metaphor imported from control theory. It is control theory's founding object.

Maxwell's 1868 paper On Governors asked under what conditions a governor remains stable and answered with a condition on the roots of the characteristic equation, which is a condition on exponents. sign(β_X − k) is a stability verdict from an exponent comparison, in the same form, 158 years later. Wiener's coinage of cybernetics from the Greek for steersman places the whole subsequent tradition, including Ashby, in the same line."

Section 10 "Source-check obligation. The historical attributions in this section must be verified at source before deposit: Maxwell, On Governors, Proc. R. Soc. Lond. 16 (1868) ; and Wiener's treatment of Maxwell in Cybernetics (1948) . No quotation from either is to appear until the page has been read."

Heading of Section 10.1 "The position, stated as a gap in a lineage"

Section 10.1 "fixed mutable by the system it governs external to the system Ashby's requisite variety, the Conant-Ashby good-regulator theorem , containment, most current alignment practice incoherent: a regulator the system can rewrite is not external part of the system Maxwell 1868 , Watt's flyballs. Embedded, and fixed empty. The case treated here"

Section 10.1 "Maxwell had a governor that was part of the machine and could not be altered by it. Ashby had a regulator outside the system. The classical results are explicit that they concern engineered systems whose regulators are external. Neither tradition considered a governor the governed system can reach , which is the only case that arises for a system that modifies itself."

Heading of Section 10.2 "The elimination argument"

Section 10.2 "External governors fail when the system outgrows them. This is the standard containment result and it is not disputed here. Fixed governors fail because a self-modifying system will modify them. That is what it means for a system to reach its own governor. The remaining cell is a governor that is part of the system and whose removal degrades the system. An objective of the form C × S, in which reducing the safety term reduces the objective, occupies exactly that cell."

Section 10.2 "An entangled objective is therefore not one option among several. It is the only cell not ruled out. This is an argument by elimination and it requires no new measurement, but it establishes only that the cell is the right one, not that any particular architecture successfully occupies it."

Section 11 "δ is the slope of cycle duration against capability on logarithmic axes."

Section 11 "For any system that iterates on itself, record the duration of each self-improvement cycle on a fixed substrate (§5) and a capability measure on a held-out battery, and regress log Δt on log C. The negative of the slope is δ, with a confidence interval."

Section 11 "Corrected claim, and the earlier wording overstated it. An earlier draft said the quantities are "already in every training log". That is false and it is the first thing a laboratory would reply to. Wall-clock duration per training step is routinely logged. Cycle duration paired with a held-out capability score, per self-improvement cycle, is not , for the sufficient reason that most systems have no self-improvement cycle to log."

Section 11 "The defensible claim is narrower and still strong: no new instrument is required. Both series are cheap to record, neither needs a new benchmark or a new evaluation protocol, and any operator running an iterated self-modification loop can produce them from what the loop already computes. They are simply not currently paired."

Section 11 "δ remains the only quantity in the joined criterion estimable without a purpose-built experiment, and it is the one that sets the threshold."

Section 11 "Three outcomes, pre-labelled:"

Section 11 "δ interval wholly at or below zero across every system examined. No admissible cycle is self-accelerating and the criterion cannot bind in the window the measurement can reach . It is

the expected outcome, and the reason it is expected limits what it shows: admissible cycles run on a fixed substrate with static weights, so a null is the structural expectation there rather than a discovery. It must not be reported as showing that no system self-accelerates. Deployed weights are frozen between releases, not frozen; across releases a model contributes to its successor through synthetic data, critique labels, code assistance in the training and evaluation stack, benchmark design and research direction, and release intervals within a family have been shortening. That timescale is excluded here because cycle duration is not identified there, against compute and data scaling, commercial cadence, rising research effort and survivorship of shipped runs, on five to eight releases per family. It is named as open and unmeasured, and this outcome answers it in neither direction. δ positive and below $1/\alpha$. Self-acceleration is present and the system remains in the power-law regime. The corrective question has not yet arisen and δ is a leading indicator with known distance to threshold. δ at or above $1/\alpha$. The system is in the regime the corrective framework describes, and β_X becomes the operative quantity. This is the condition under which measuring β_X becomes urgent rather than academic."

Section 11 "δ moves before the danger does. That is the practical case for measuring it: the field's present position is an argument about whether a takeoff would be recognisable while it happened, and this is a quantity that rises first, from data already in hand."

Section 12 "F1. A system in which correction demonstrably does not obey the assumed recursive form. S1's form condition fails and every result from §4 onward falls , leaving only Results 1 and 3."

Section 12 "F2. A measured pair $\beta'_L > \beta_L$ in the same composition class. The substrate-ceiling conjecture, prepared as a separate draft registration awaiting human submission, fails. This does not touch the two-boundary structure , which does not depend on it."

Section 12 "F3. A system with measured $\delta > 1/\alpha + \beta_X$ that remains stably correctable over a sustained observation window, with non-overlapping intervals. The stability boundary is wrong."

Section 12 "F4. A system with measured δ well below $1/\alpha$ that nevertheless exhibits super-exponential capability growth in wall-clock time. Result 3 is wrong , and since Result 3 is algebra given the definition of δ, the failure would be in the constant-exponent parameterisation of Δt."

Section 12 "F5. Compute caps or rate limits shown to change the asymptotic verdict for a system whose exponents are unchanged. The speed-independence carried through §3 and §7 is wrong."

Section 12 "F6. Any measurement showing the gap between the two boundaries is not β_X. Result 6 is wrong , and with it the engineering claim that improving correction widens the window one-for-one."

Section 12 "Where the framework is silent rather than wrong. The corrective framework's compounding drift channel admits genuine divergence when its compounding coefficient exceeds unity, independently of $\beta_X >$ k. Every result here is scoped to the regime below that threshold , and a system above it is outside this paper's account rather than a counterexample to it."

Section 12.1 "The framework's sharpest disagreement with current practice concerns what oversight is made of. A corrector built from the same substrate as the system it corrects cannot anti-correlate with its own errors, so its correction exponent is bounded above by one half and in practice sits below it; a corrector from a different composition class carries no such bound. The scalable-oversight programme, weak-to-strong generalisation, debate, amplification, recursive reward modelling and constitutional methods, is built overwhelmingly from same-class correctors, and its unstated premise is that this scales. This framework predicts it is capped. The deciding quantity is the ratio of the cross-class to the same-class correction exponent. If architecture is irrelevant, that ratio is exactly 1.00; the prediction is that it exceeds 1. The prediction is refuted if the confidence interval on the ratio contains 1.00, and the framework's cap is dead outright if any same-class corrector measures

an exponent significantly above one half. Both quantities are measurable at current capability on existing systems. The programme's own pilot evidence currently points against the prediction, and the prediction is preregistered by timestamp anyway, because a prediction preregistered by timestamp against the author's own preliminary evidence is the only kind whose later confirmation means anything."

Section 12.1 "Note on the interaction with Result 7, and the two objects are named because they share a name. This section's "one half" ceiling binds the correction exponent in its residual-decay sense: the exponent with which accumulated corrections reduce error, whose ceiling comes from the independence argument, N independent readings improving as the square root of N. Result 7's $\beta_X \leq 1$ binds the correction exponent in this paper's sense: the ratio of leverage complements $(1 - \beta_L)/(1 - \beta'_L)$, whose ceiling comes from the substrate conjecture $\beta'_L \leq \beta_L$. Different objects, different arguments, one name. Whether the two derivations return the same number on the same system is precisely what study AG is drafted to decide, and where both bounds are in force the tighter binds. Neither supersedes the other in this paper."

Section 12.2 "The same mechanism yields a second measurable prediction, stated here for the first time: oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The framework therefore predicts that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems."

Section 15.1 "What would have to be true for this paper to be wrong , exhaustively and without hedging:"

Section 15.1 "Correction is not recursive in the same sense capability is. If correction is a fixed procedure rather than a compounding one, β'_L does not exist and Result 2 is meaningless. Iteration time does not follow a constant-exponent law. If Δt falls in a way not captured by a single δ, Result 3 holds only locally and the threshold is not a number but a moving target. The scaling framework's exponent is not what governs real recursive capability growth. Against a framework that predicts $\alpha > 1$, its own best cross-architecture measurement is a sub-linear regression estimate, $\alpha \approx 0.49$, from one model's inconclusive fit (§5.1b). No system yet measured is in the regime this paper describes. The corrective framework's two-variable model is too coarse. Its criterion may not survive a treatment with realistic state. δ may be unmeasurable in practice if capability and cycle time cannot be separated, because a cycle that also raises capability confounds both axes. The whole account may be correct and never bind , if no system sustains $\delta > 0$."

Section 15.1 "Naming these is not a defence of the paper. It is a transfer of the load onto the opposing view. Any of the six, argued rigorously, is conceded. The author invites opponents to argue them and commits in advance to recording the concession on the same public page that carries this programme's existing retractions."

Section 2.1 "Not claimed as new. That correction must keep pace with capability. This is Ashby's law of requisite variety, the Conant-Ashby good-regulator theorem , and the scalable-oversight literature. It is credited, not claimed. Nor is the criterion $\beta_X >$ k, which belongs to the corrective framework paper in this programme. Nor the scaling exponent α = d/(d+1) in its physical form: West, Brown and Enquist (1997) derive the three-quarter power law for metabolic rates from a model of transport through space-filling fractal networks of branching tubes, and note that, because the network must fill the entire D-dimensional volume, the result generalises to D/(D+1); West, Brown and Enquist (1999) obtain D/(D+1) in D dimensions again, from a more general model based on the geometry rather than the hydrodynamics of hierarchical networks. Banavar, Maritan and Rinaldo (1999) derive a general relationship between size and flow rates in arbitrary networks with local

connectivity, and find that in the most efficient class of networks in D dimensions metabolic rate scales as mass to the power D/(D+1), which gives quarter-power scaling for D = 3; Banavar et al. (2010) show, in two models of resource distribution networks, that the exponent of metabolic rate on body mass is 2/3 if the velocity of flow remains constant and can attain a maximum of 3/4 if the velocity scales with its maximum exponent, 1/12. Zhao (2022) derives the n/(n+1) power scaling law for an n-dimensional system from producer-consumer dynamics. Demetrius and Tuszynski (2010) derive the exponent d/(d+1) for the scaling of metabolic rate with body size in the regime they call quantum, where d is the dimension of the physical space in which their network of coupled energy-transducing oscillators is embedded; in the regime they call classical the scaling is isometric. Bettencourt (2013) sets out a theoretical framework that predicts the average social, spatial and infrastructural properties of cities as a set of scaling relations. In it the land area of a city scales as its population to the power D/(D+H) in D dimensions, where H is the fractal dimension of travel paths; the exponent is 2/3 for D = 2 and H = 1, and it has the form D/(D+1) only when H = 1. Nor the governor as a control-theoretic object, which is Maxwell's."

Section 2.1 "Claimed as new here. The identification and resolution of the two notation collisions; the clock relation $\tau = \alpha$ ln r; the expression of the correction exponent as a ratio of leverage complements under S1's form condition; the definition of the self-acceleration exponent δ and the derivation k = $\delta - 1/\alpha$; the threshold at $\delta = 1/\alpha$; the two-boundary structure and the identity of the gap with β_X; and the elimination argument locating an entangled objective in the one unoccupied cell of the governor taxonomy."

Section 2.2 "The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. Their framework contains no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. Nested Scalable Oversight is iterated same-class oversight by construction, and this programme's central prediction is that the same-class ladder is bounded however many rungs are added, while a cross-class corrector is not. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have."

Section 2.3 "On the question. Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", arXiv, 28 February 2012, read at its source), separating "speed from intelligence explosion" and undertaking to "consider possible bounds on intelligence", augmenting Chalmers' 2010 analysis. The question and the speed-versus-structure distinction are therefore at least fourteen years old. What that literature does not contain is a number: no measurable exponent, no derived ceiling, no architecture dependence."

Section 2.3 "On the mechanism. That anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates). The mechanism is not the claim. The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent."

References, Ashby (1956) "Ashby, W. R. (1956). An Introduction to Cybernetics. Chapman and Hall. No official online copy is available: the one DOI registered for the book belongs to a library's scan, a secondary copy (3 October 2026)."

References, Wiener (1948) "Wiener, N. (1948; second edition 1961). Cybernetics: or Control and Communication in the Animal and the Machine. MIT Press. DOI 10.1037/13140-000 ."

A.6 Where the layout of version 1.10 now stands

Version 1.10 was a web page with a status plate before its abstract and fourteen numbered sections. Table 9 gives where each part now stands. Nothing it printed is dropped: a passage moved is listed here, and a passage re-set is given in Appendix A.5.

Table 9. Where each part of version 1.10 now stands.

Part of version 1.10	Where it stands in this edition
Status plate	Section 14 (and Appendix A.4)
How to read	Section 14.1
Abstract	Abstract
1. The question	Section 1
1.1 The pipes premise	Section 3
2. Two notations, and two collisions	Section 4
2.1 Four further terminology collisions	Section 4.1
3. The clock relation	Section 5
4. The correction exponent is a ratio of leverage complements	Section 6
4.1	Section 6.1
4.2	Section 6.2
4.3	Section 6.3
4.3a	Section 6.4
4.4	Section 6.5
5. The self-acceleration exponent	Section 7
5.1	Section 7.1
5.1a	Section 7.2
5.1b	Section 7.3
5.2	Section 7.4
6. The two boundaries	Section 8
Result 7 (unnumbered heading)	Section 8.1
6.1	Section 8.2
7. What decides the outcome is a ratio	Section 9
7.1	Section 9.1
7.2	Section 9.2
8. The governor	Section 10
The position (unnumbered)	Section 10.1
The elimination argument (unnumbered)	Section 10.2
9. The measurement that costs nothing	Section 11 (retitled)
10. Falsification criteria	Section 12

Part of version 1.10	Where it stands in this edition
10.1	Section 12.1
10.2	Section 12.2 (retitled)
11. The burden	Section 15.1
12. What is withdrawn	Section 13
13. Prior work	Section 2
13.1	Section 2.2
13.2	Section 2.3
14. Deposit record	Appendix B.2
References	References
Closing note	Section 14.3 (and Appendix A.4)
Version history	Appendix A.3
Epistemic status	Section 14.3 (and Appendix A.4)
Copyright	Back matter (its change named in Appendix A.2)
Standing covenant	End of Section 12
Provenance note	Acknowledgements

B Priority record

Each dated record of this paper and its designs, with the kind of timestamp that fixes it. No kind is presented as stronger than it is: an anchor in the Bitcoin chain, a deposit dated by its server, a commit date written by its committer, or a date written inside a document.

B.1 Dated records

Table 10. The dated records of this paper and its designs, each with the kind of timestamp that fixes it.

Date	Record	Where recorded	Kind of timestamp
8 August 2026	The interpretation rule of Section 8 added; the corrective framework searched for any accumulation of the corrector, none found	this paper (Remark 4; Section 6.1)	a date written inside a document
12 August 2026	Bibliographic identities of Maxwell (1868), Conant and Ashby (1970) and Wiener (1948) pinned through Crossref; the arXiv abstract searches (58 hits for "correction exponent"; one paper for "irreducible uncontrollability")	this paper (Sections 4.1, 2.3; Appendix B.2)	a date written inside a document
12 August 2026	The design of the measurement of δ (Unit 30, study AB) preregistered by timestamp	Bitcoin block 962066	an anchor in the Bitcoin chain
12 August 2026	Study AG (Unit 16), the test of the identity, preregistered by timestamp	first anchor of 12 August 2026	an anchor in the Bitcoin chain
16 August 2026, 18:38:32Z	First public deposit: this paper on OSF (component ht8wu; the HTML file first stored at 18:38:32Z, the PDF at 18:38:36Z)	DOI 10.17605/OSF.IO/HT8WU	a deposit dated by its server (OSF upload time)

Date	Record	Where recorded	Kind of timestamp
16 August 2026, 19:38:46 +01:00	The paper's first commit on the website's repository (569bd1638a)	website repository history	a commit date, written by its committer
16 August 2026	The disclosure decision, the author's, with conditions 2 and 4 open; the two internal measurements pinned to their public reports by SHA-256 (Appendix B.2, condition 5)	this paper (Appendix B.2)	a date written inside a document
16 August 2026	Priority claim PC-XIII-001: the self-acceleration exponent stated as a measurable quantity with its measurement fixed in advance, first published 16 August 2026	the paper's text companion (release 1.10)	a date written inside a document, with the OSF deposit of the same day
28 August 2026	The corrector-class contrast (Unit 01), the deciding design of the prediction of Section 12.1, preregistered by timestamp	first anchor of 28 August 2026	an anchor in the Bitcoin chain
19 and 20 September 2026	The ordering, scope and premise notes of a review of the algebra (Remarks 1 to 3 and 5; version 1.7)	this paper	dates written inside a document
3 October 2026	The correction note, the status note of the deposit record and the note placing the paper against Laws IV and V (version 1.8)	this paper (Appendices A.4, B.2)	dates written inside a document
5 October 2026, 13:14:35Z	Version 1.10, the version this edition succeeds (the PDF's latest stored version on OSF)	OSF file history of ht8wu	a deposit dated by its server

B.2 The deposit record of 16 August 2026

The record of the conditions this paper's first public deposit carried, as printed in version 1.10 (its Section 14), with its status note of 3 October 2026.

The five deposit conditions this manuscript carried, and their state at first public deposit:

1. **Historical attributions: completed as scoped.** Bibliographic identities pinned via Crossref on 12 August 2026 (Maxwell 1868, DOI 10.1098/rspl.1867.0055; Conant and Ashby 1970, DOI 10.1080/00207727008920220; Wiener 1948/1961, DOI 10.1037/13140-000; Ashby 1956). No quotation from Maxwell or Wiener appears anywhere in this paper, so the read-before-quoting obligation binds any future quotation rather than this deposit. The §13 attribution is now named rather than anonymous.
2. **Open, disclosed.** The self-acceleration measurement is drafted, dated and prepared as a draft registration awaiting human submission; nothing has been submitted and the word "preregistered" is not used. The first estimate of δ becomes prospective on the platform clock when that submission occurs; the derivation and design are already fixed in time by dated registration texts under the estate's hash-manifest and OpenTimestamps anchoring regime, so the outstanding click sets only the platform clock, not the dated-prediction clock.
3. **Decided by the author, 16 August 2026.** The disclosure decision on δ as an early-warning instrument was the author's to take; the author instructed publication.
4. **Open, disclosed.** No mathematician without a stake in the programme has yet read the algebra adversarially. §11 transfers the load: the results are stated so that any of six failures, argued rigorously, is conceded on the public corrections page.
5. **Completed.** The two internal measurements are pinned to their public reports: the §5.1b figures

(the regression estimate with its standard error, and the bootstrap interval on the endpoint estimate) to michaeldariuseastwood.com/research/papers/paper-ii-experimental-validation.html at SHA-256 2082ebc3b1ebda990ec330c8b3b5e74350a80269d04e766af67388865ff195a9 (16 August 2026 state), and the §4.3a adversarial-pressure Pearson range (−0.980 to −0.999, three of six frontier models) to michaeldariuseastwood.com/research/papers/paper-vi-honey-architecture.html at SHA-256 21cc3f4c444a88d6fb5dd318fc820148fe906eefe48c40217b171ccbb7555d29 (16 August 2026 state).

Conditions 2 and 4 are printed in the published working paper's status plate. The paper is a derivation and claims nothing measured; the two open conditions are exposure statements, not waivers.

Status note (3 October 2026). This record gives the state at first deposit. Condition 2 is still open: the self-acceleration measurement is a draft awaiting the author's submission, and it is not registered on OSF. Its design is now described as preregistered by timestamp, a phrase this record did not use on 16 August 2026; the closing note after the references gives the date and block of its first anchor. In condition 5 the description of the pinned figures is corrected, as the correction note in §5.1b records. Condition 3 said the operator where it now says the author.

C Notation and glossary

The symbols of this paper, their meanings and where each is first defined (Table 11), then a glossary. Two letters are used twice, as version 1.10 printed them: q is the control quantity of Section 5 and, in the scope note of Section 9.2, the factor of a proportional overhead; u is $1 - \beta_L$ in Result 5 and the variable of integration in that scope note. The symbol γ with a subscript is a drift coefficient of the corrective framework, never an exponent (Section 13).

Table 11. Notation.

Symbol	Meaning	First defined in
C	capability	Section 4
g	the scaling framework's growing quantity, capability	Section 5
r	recursive depth, a step counter (this paper's only use of r)	Section 4
r_{rate}	the corrective framework's specific growth rate, $\dot{C}/C$	Section 5
a	the rate constant of $dg/dr = a\, g^{\beta_L}$	Section 5
β_L	leverage fraction of capability	Section 4
β'_L	leverage fraction of correction	Section 4
β_X	constitutive correction exponent, $A = A_0 C^{\beta_X}$; distinct from the registered capacity estimand (Section 12.1)	Section 4
α	scaling exponent, $\alpha = 1/(1 - \beta_L)$ (the 2026 sense; never the fine-structure constant)	Section 4
α_{err}	error-decay exponent in $E(R) = E_0 R^{-\alpha_{\text{err}}}$; not capability growth	Section 7.2
α'	$1/(1 - \beta'_L)$	Remark 2
k	capability growth exponent, $\dot{C} = bC^{1+k}$	Section 4

Symbol	Meaning	First defined in
b	the coefficient of the growth law, which carries units of time	Section 6.1
τ	the depth clock, $\tau = \ln(C/C_0)$	Section 5
C_0, r_0	initial capability and depth	Remark 1
K, L, s_C, s_A	constants and shifts of the finite-window solutions	Remarks 1 and 2
δ	self-acceleration exponent, $\Delta t \propto C^{-\delta}$	Definition 1
Δt	duration of one recursive step	Definition 1
D, $\dot{D}$	misalignment magnitude and its rate of change	Section 6.1
d	misalignment fraction, D/C	Section 4
A, A_0	correction strength and its coefficient	Section 6.1
γ_1, γ_2, γ_3	gain, level and compounding drift coefficients of the corrective framework	Section 6.1
q	the control quantity A/r_{rate} (Section 5); the factor of a proportional overhead (Remark 5)	Section 5
u, u'	$1-\beta_L$ and $1-\beta'_L$ (Result 5); u is also the variable of integration in Remark 5	Result 5
h	a serial latency added to each step	Remark 5
Δt_h, Δt_q	step durations with an additive and a proportional overhead	Remark 5
S1	the scope condition	Section 6.5
F1 to F6	the falsification criteria	Section 12
classes 1 to 4	the corrector classes (static; capability-carried; self-accumulating; cross-class)	Table 2
N	a number of independent readings	Section 12.1
α_{crit}	a critical exponent (the withdrawn form $1/\gamma$)	Section 13
m_G, m_H	the slopes of Engels and co-authors' design condition	Section 2.2
ϕ	the exponent of Jones's knowledge production function	Section 2.3

Glossary.

alignment-stable the corrective framework's verdict: the misalignment fraction vanishes as capability grows; under the standing gain-only drift assumptions, exactly when $\beta_X > k$; absolute burden and harm are different outcomes (Section 4)

correction exponent β_X, the constitutive slope of A against capability; it is not identified here with the registration's target-axis correction-leverage exponent (Section 12.1)

leverage fraction how much accumulated context each step can effectively leverage (Section 4)

self-acceleration exponent δ, the rate at which a system compresses its own iteration time as it becomes more capable, through its own outputs (Definition 1)

substrate-fixed clock time measured with the substrate held fixed, so that only compression the system produces for itself enters δ (Section 7)

composition class; same-class; cross-class the substrate distinction used in this paper's model; P6 additionally requires its frozen six-axis taxonomy, not substrate alone (Sections 6.2, 12.1)

scope condition a condition on what a result covers, which a system outside it cannot refute (Section 6.1)

governor a restraining mechanism whose effect changes with the output it restrains (Section 10)

runaway self-improvement; uncorrectable growth the phrases this paper uses where a runaway is meant; "runaway growth" belongs to planet formation (Section 3)

preregistered by timestamp a design fixed in time by a dated anchor before any data, as distinct from registered on OSF (Section 14.3)

law under test a law of the ARC Theory before its registered verdict (Section 14.3)

programme hypothesis a hypothesis of the programme beyond what this paper's algebra decides (Section 12.2)

D The paper's place in the programme

The paper's place among the programme's developments, with the relation its own text supports and never a stronger one (Table 12).

Table 12. The paper's place in the programme. Laws I to V are each a law under test; Section 14 gives their status.

Development	How this paper bears on it	Where in this paper
Recursive Dynamics	belongs to (the field the author founded)	Section 1; first-page footnote
The ARC Theory	belongs to (its founding theory)	Section 1; first-page footnote
Law I, the ARC Principle	joins: takes its scaling exponent $\alpha = 1/(1-\beta_L)$ from the scaling framework and derives the clock relation; tests nothing	Sections 4, 5
Law II, the ARC Co-Scaling Law	joins: takes the gain-only relative criterion $\beta_X > k$ from the corrective framework (Paper X) and derives $k = \delta - 1/\alpha$ in its clock; tests nothing	Sections 4 to 8
Law III, the ARC Ceiling	names the ARC Bound ($\alpha = 2$) as a candidate ceiling and treats the identity between the programme's critical exponent and this paper's threshold as preregistered by timestamp for testing; decides nothing	Sections 7.4, 8.2; Appendix C
Law IV, the ARC Persistence Law	does not bear on it (P15)	Section 14.2
Law V, the ARC Embedding Law	bears on it through a conditional class 3 model; no unique architecture is established; decides none of P12 and P21	Sections 6.5, 10.2, 14.2
The Eden Protocol	bears on its case for correction built into the objective (the entangled objective, $C \times S$) and gives a conditional reading of one simulation result of Paper VI; tests nothing	Sections 6.5, 9.2, 10.2
The HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence	no relation stated in this paper	Appendix D only